



SCIENTIFIC DATA VISUALIZATION

AGENDA

Introduction

Basics of data visualization

Basic guidelines and common pitfalls

 Break

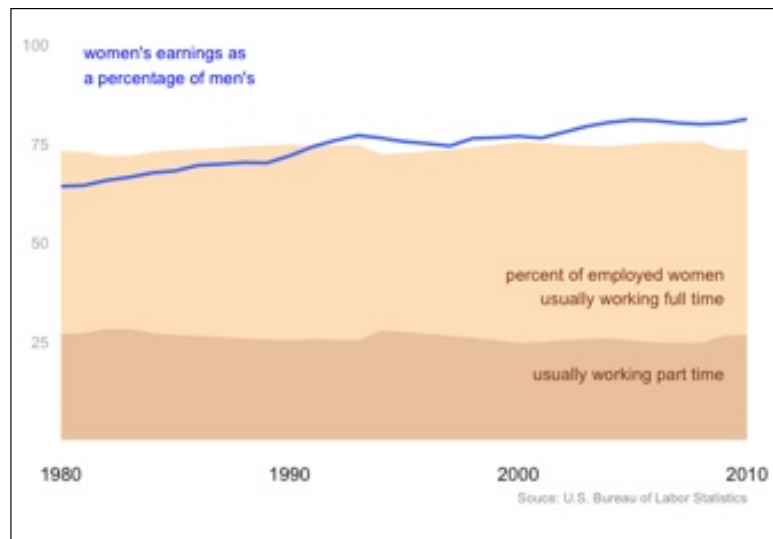
Examples and exercises in ggplot2

GEORGIOS KARAMANIS

Psychiatrist, Gender identity clinic, Uppsala University Hospital

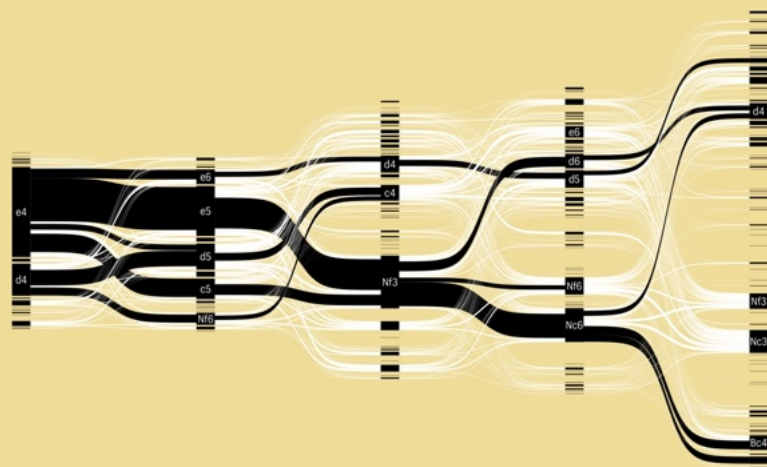
PhD student, Institution for medical sciences, Clinical Psychiatry, Uppsala University

Data visualization designer, freelancer

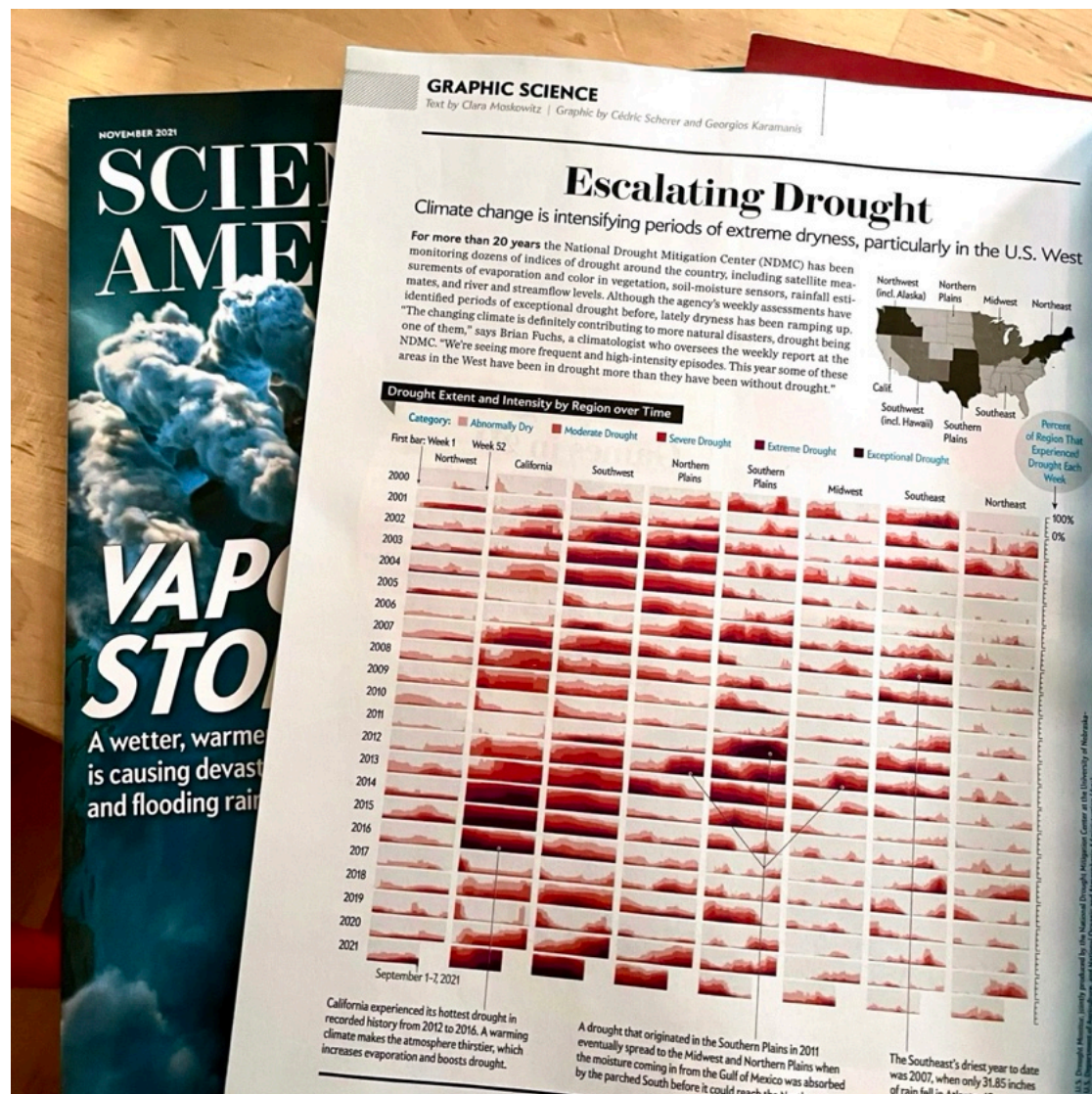


First Five Moves

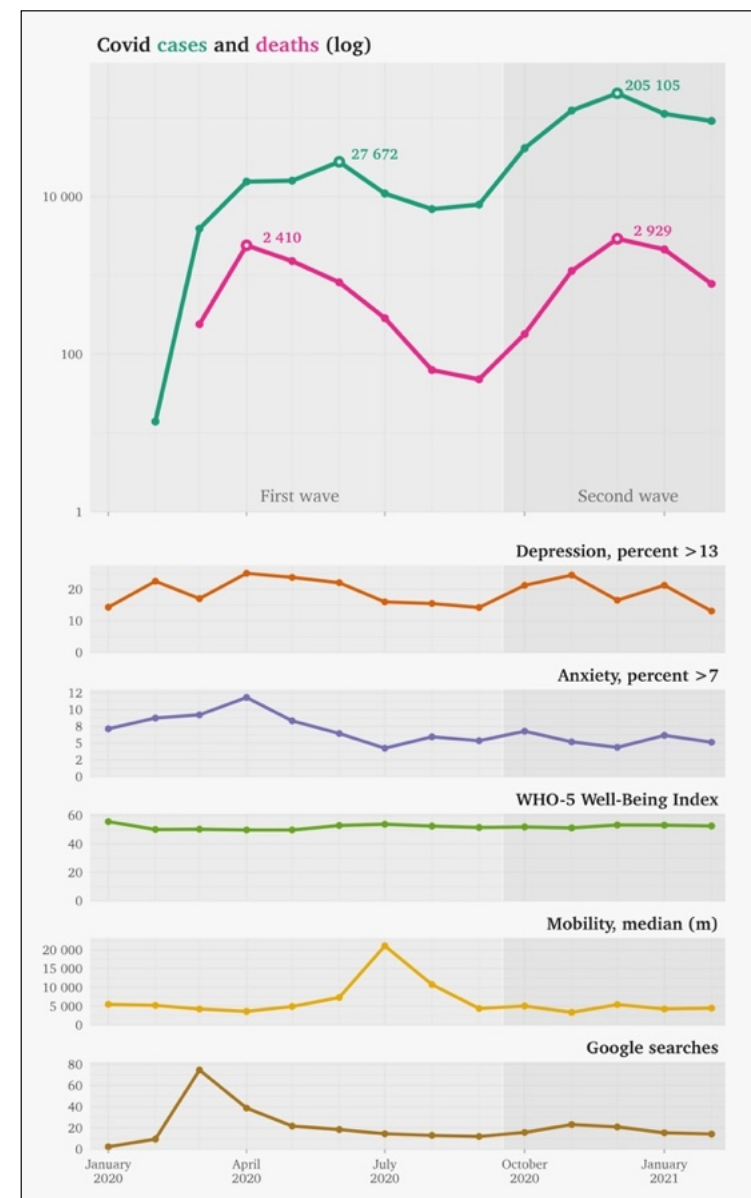
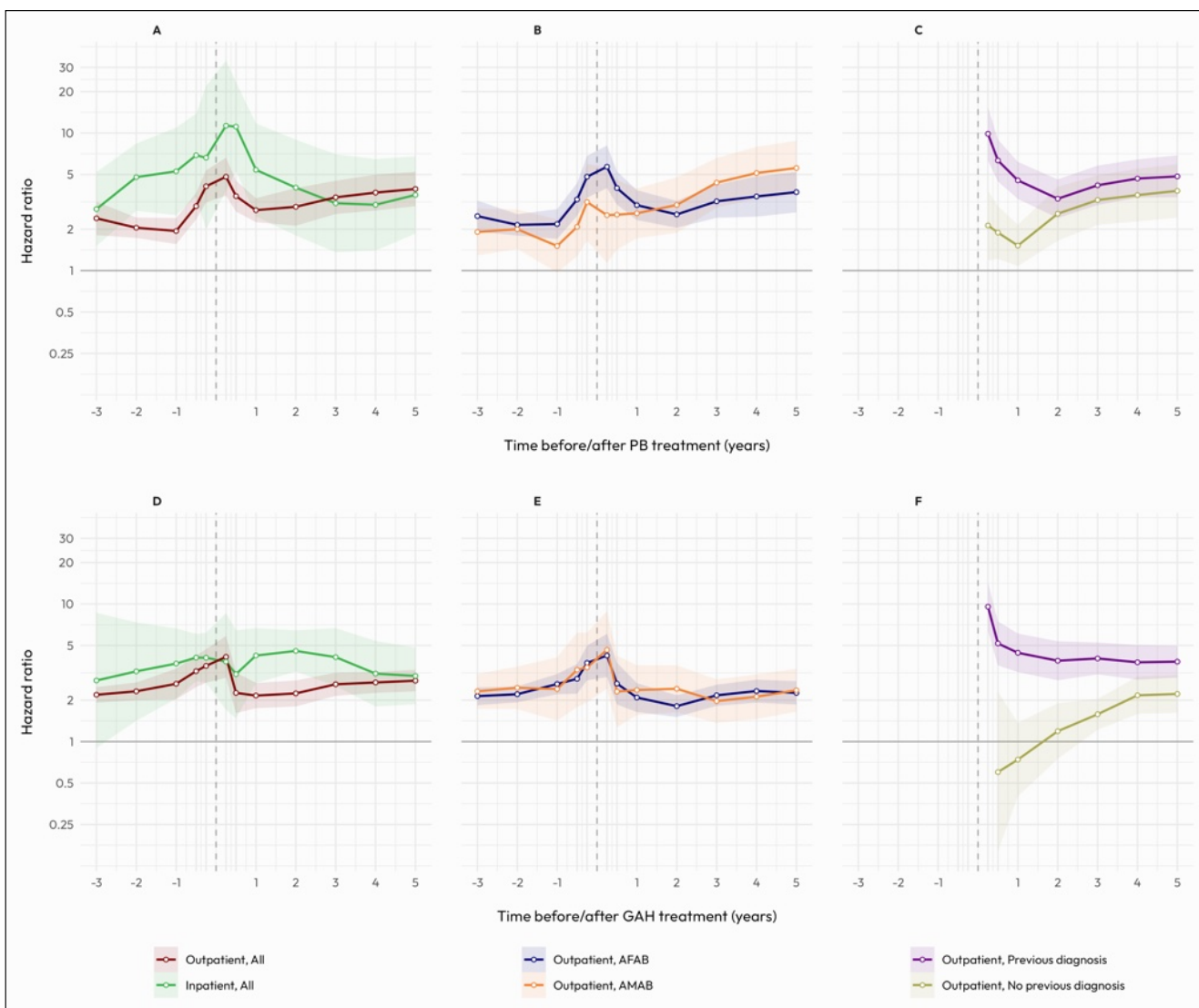
Flow of opening moves in over 20 000 chess games from Lichess.org. The most popular moves are highlighted in black, revealing common opening strategies.



Source: Lichess.org via Kaggle/Mitchell J. · Graphic: Georgios Karamanis



<https://www.scientificamerican.com/article/climate-change-drives-escalating-drought/#>



INTRODUCTION

DATA VISUALIZATION

Data visualization blends art and science to convey stories from data via graphical representations

Controversy of art posts on r/pics
posts featuring **women** posing with their art are more likely to be downvoted than posts with **men** posing

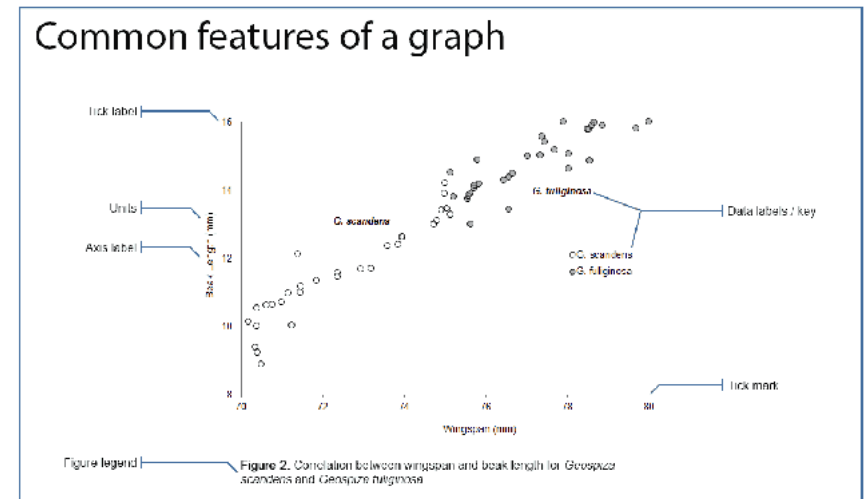


<https://erdavis.com/2021/06/14/do-women-who-pose-with-their-art-on-reddit-get-more-upvotes/>

SCIENTIFIC CHARTS

“Figures in scientific publications are critically important because they often show the data supporting key findings”

Weissgerber TL, Milic NM, Winham SJ, Garovic VD (2015) Beyond Bar and Line Graphs: Time for a New Data Presentation Paradigm. PLoS Biol 13(4): e1002128. <https://doi.org/10.1371/journal.pbio.1002128>

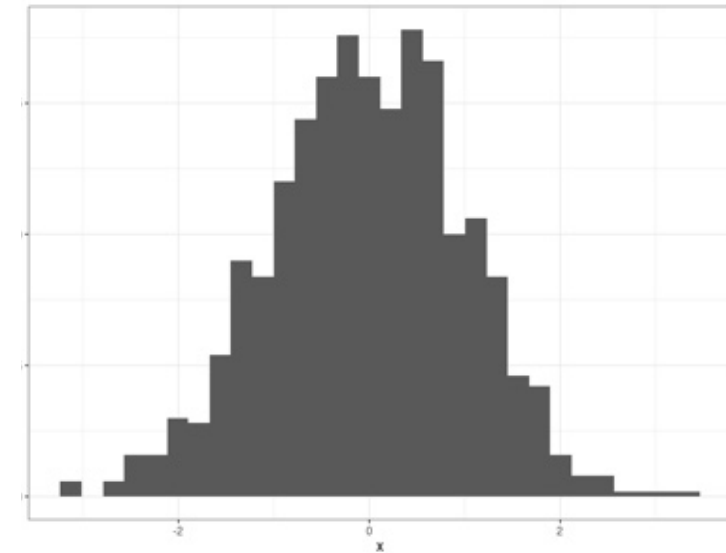


<https://www.clips.edu.au/displaying-data/>

WHY CHARTS?

Easier overview of data
(especially complex)

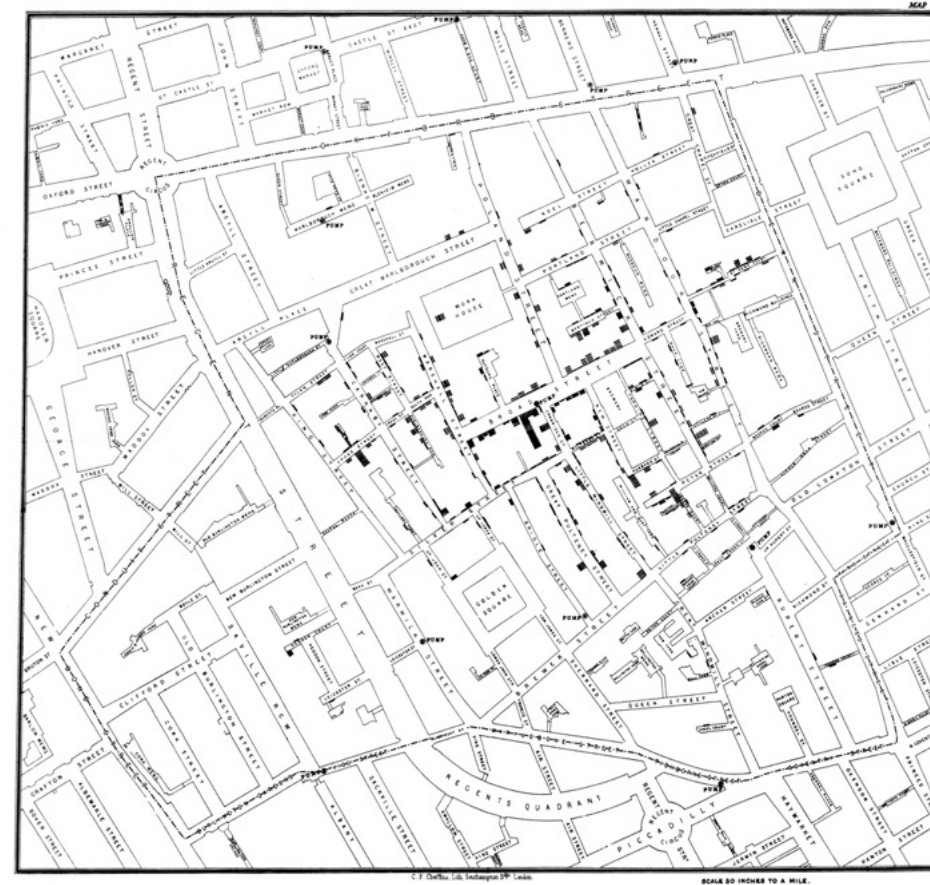
	x
1	0.376935655
2	-0.603182135
3	0.659734512
4	-1.597222895
5	-0.645683665
6	-0.929817315
7	1.401684999
8	0.520906260
9	0.369003441
10	0.573936602
11	1.304568093
12	-2.537849064
13	0.920903182
14	1.419055702
15	-0.288114527
16	1.842139177
17	0.647224489
18	1.793445537
19	-2.002214213
20	0.885130196
21	-0.369039857
22	-0.126587806



WHY CHARTS?

Find new patterns

BROAD STREET PUMP OUTBREAK, JOHN SNOW, 1854

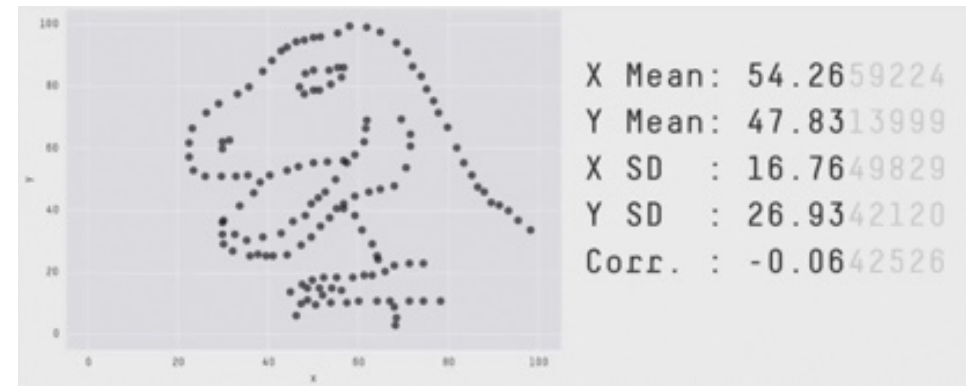


https://en.wikipedia.org/wiki/1854_Broad_Street_cholera_outbreak

WHY CHARTS?

Easier for comparisons of data and their statistics

THE DATASAUROUS DOZEN

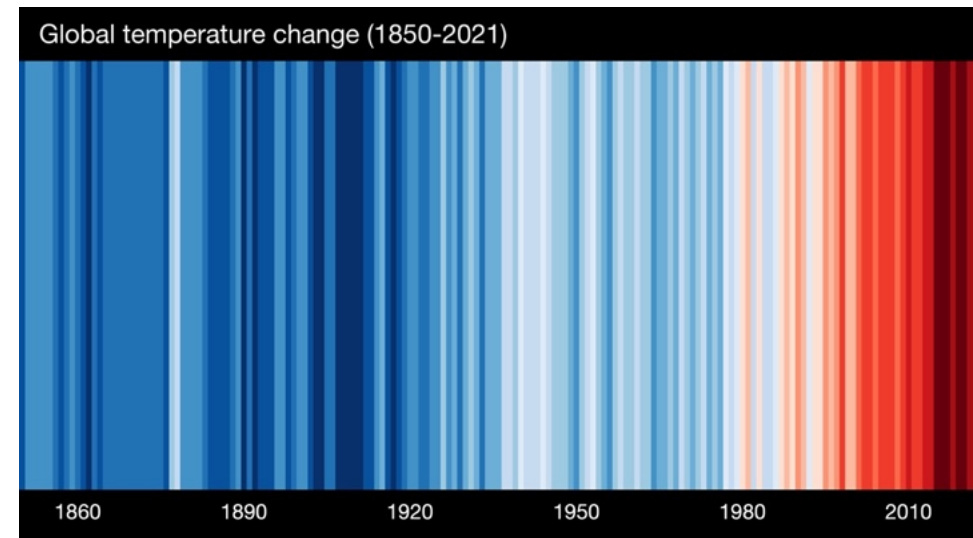


<https://www.autodesk.com/research/publications/same-stats-different-graphs>

WHY CHARTS?

To better communicate your point

WARMING STRIPES, ED HAWKINGS, 2018

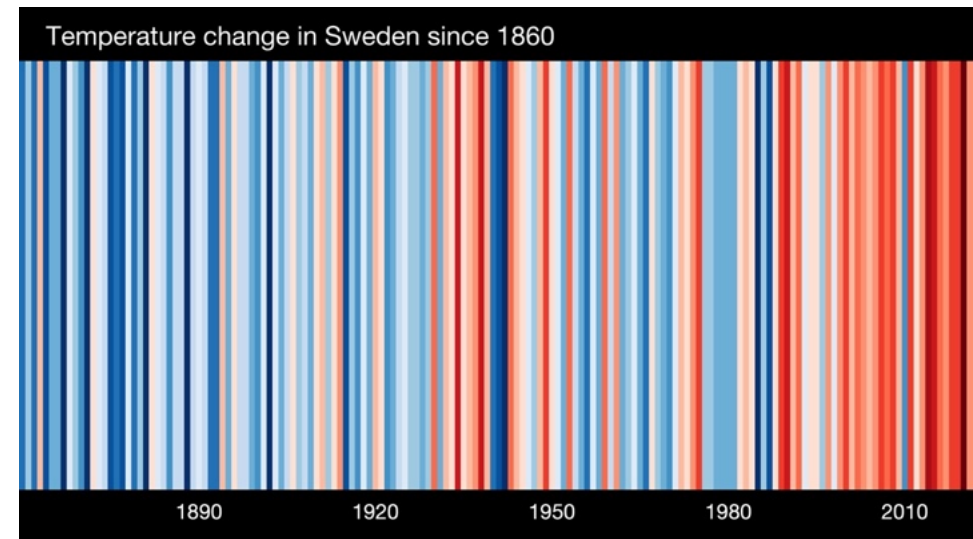


<https://www.climate-lab-book.ac.uk/2018/warming-stripes/>

WHY CHARTS?

To better communicate your point

WARMING STRIPES, ED HAWKINGS, 2018

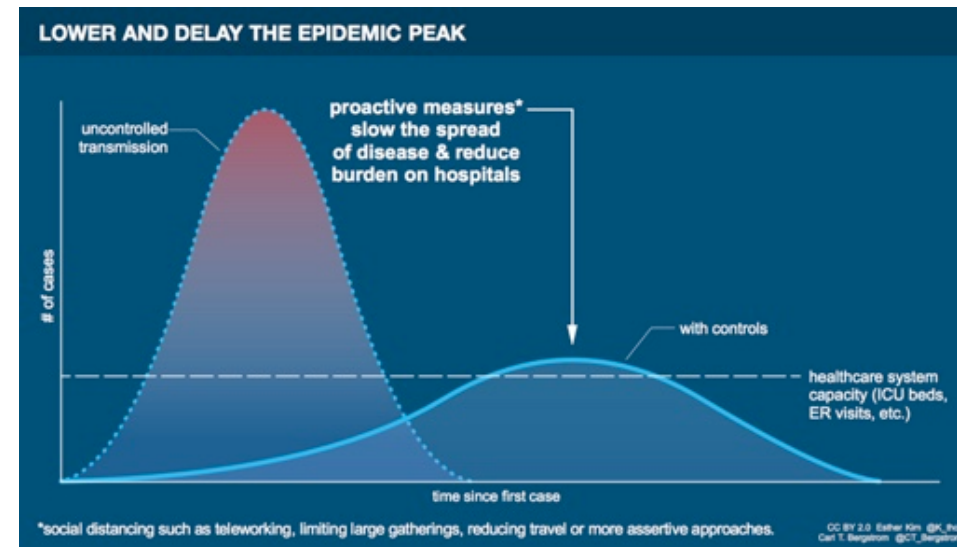


<https://showyourstripes.info/l/europe/sweden/all>

WHY CHARTS?

They look good, are memorable and shareable

ESTHER KIM & CARL T. BERGSTROM, 2019

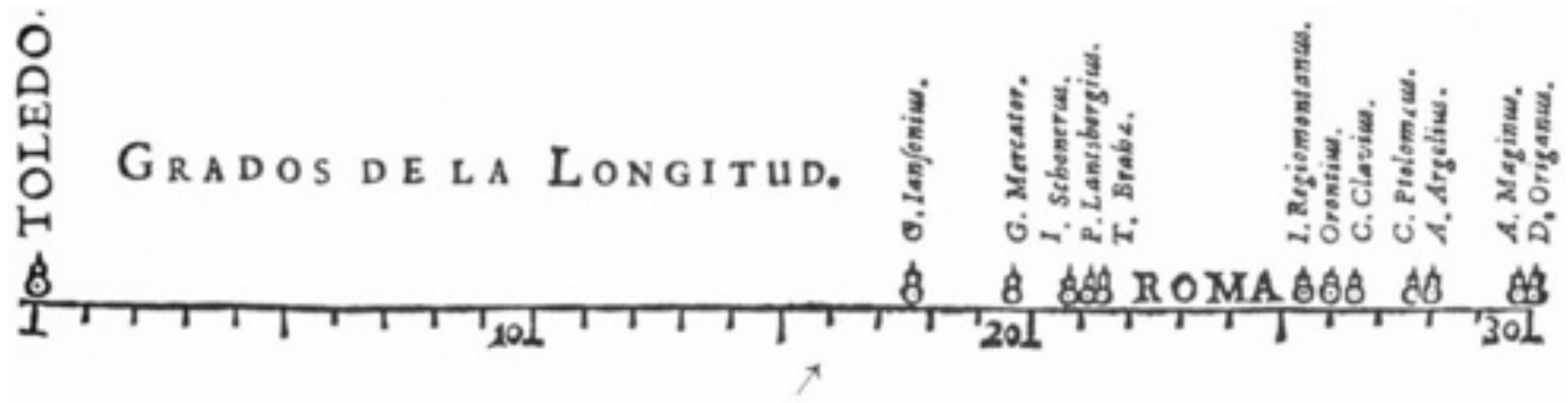


<http://ctbergstrom.com/covid19.html>



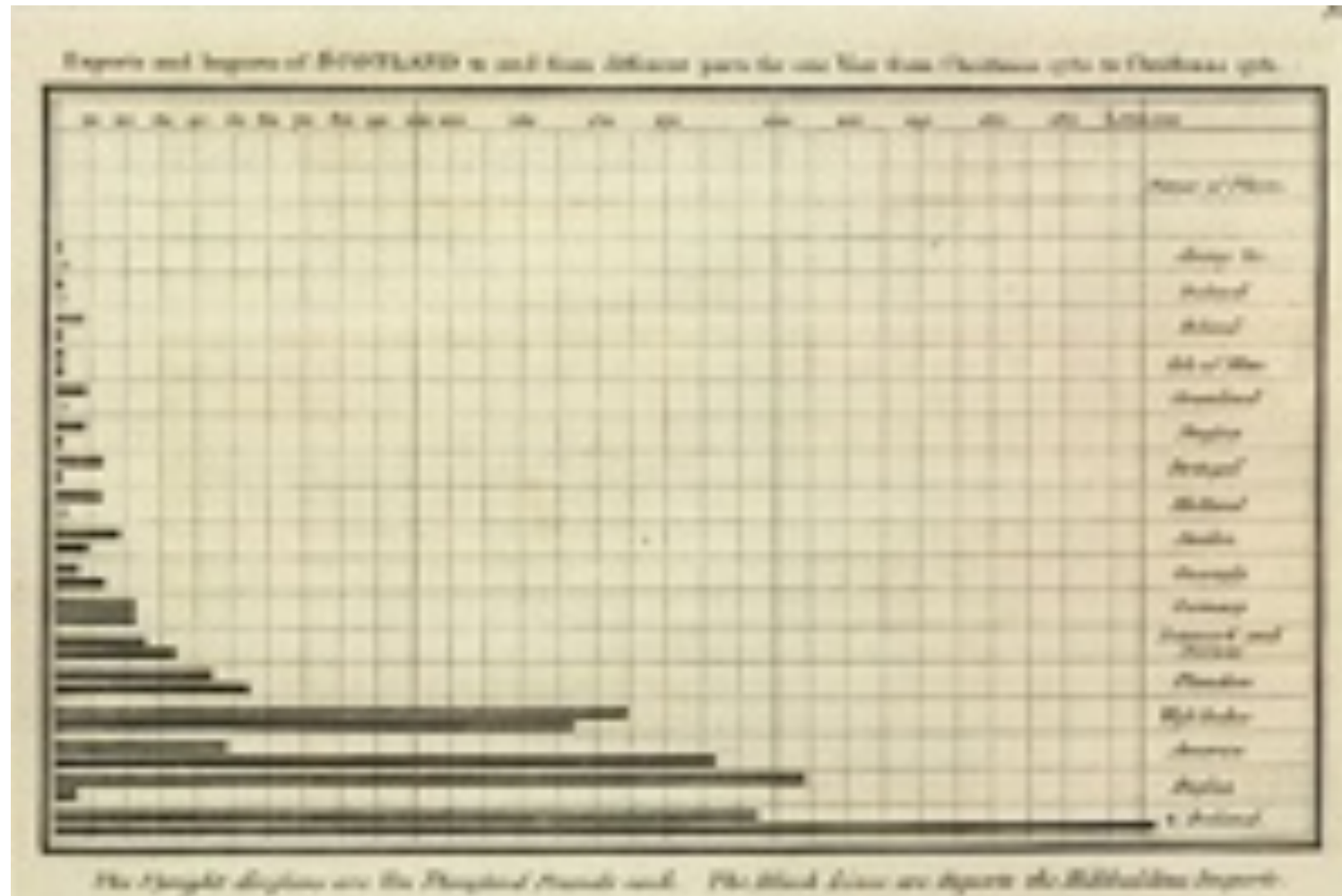
**SCIENTIFIC CHARTS
HAVE A LONG HISTORY**

MICHAEL FLORENT VAN LANGREN, 1644



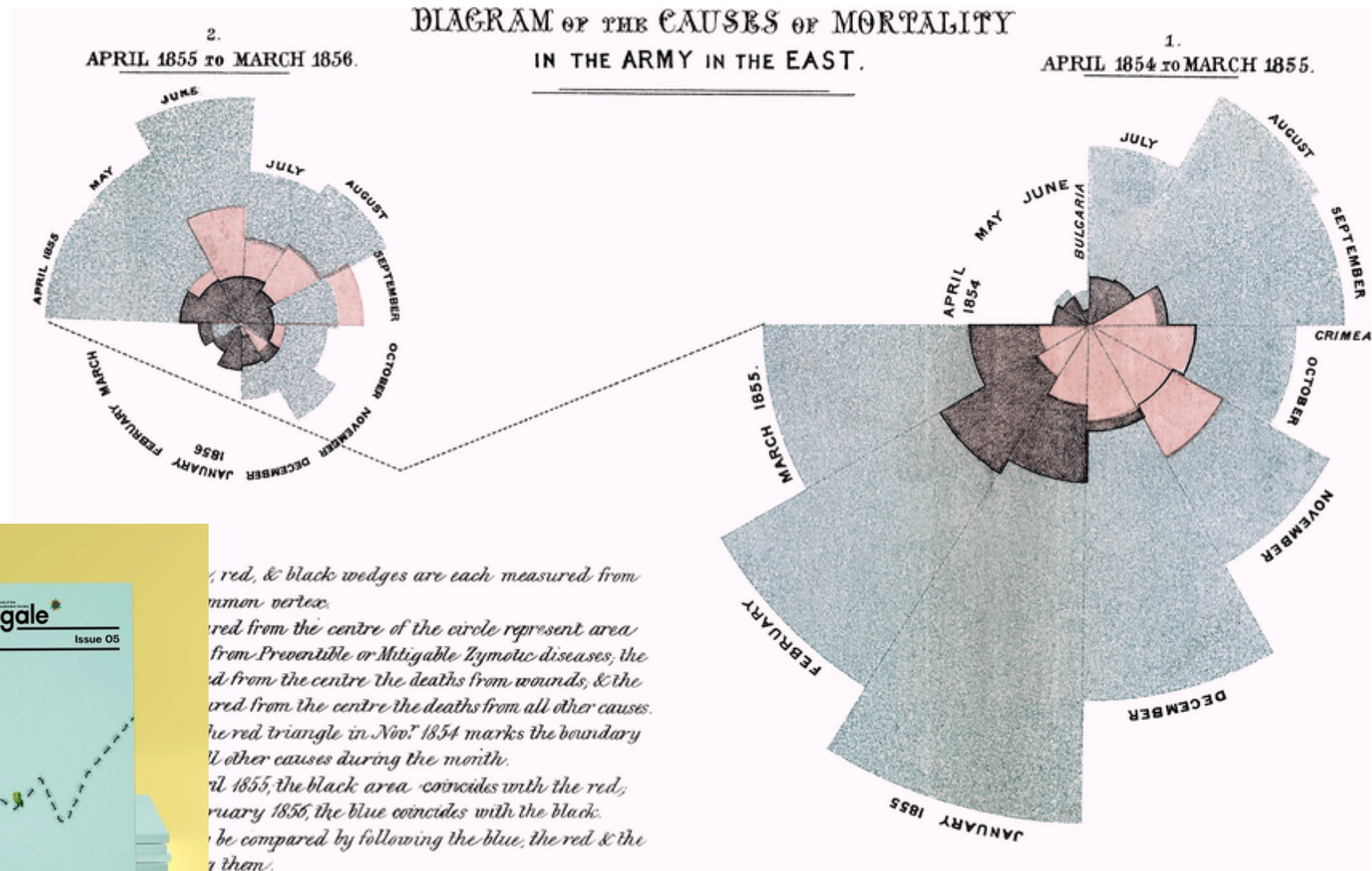
<https://historyofinformation.com/detail.php?id=3415>

WILLIAM PLAYFAIR, 1786



<https://data-xtractor.com/blog/data-visualization/chart-history-who-invented-this-chart-type/>

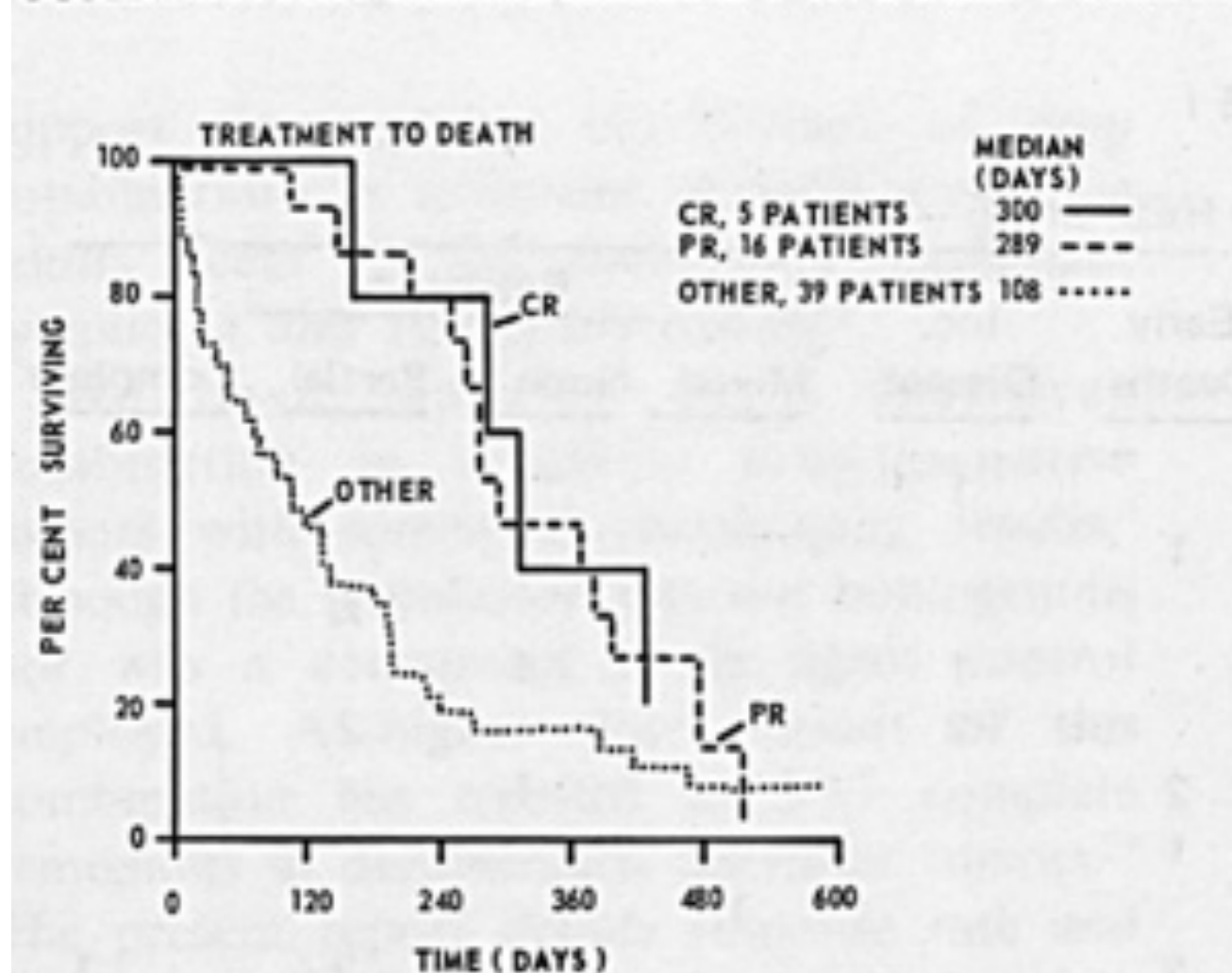
DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST, FLORENCE NIGHTINGALE, 1858



https://en.wikipedia.org/wiki/Florence_Nightingale

EDWARD GEHAN, 1971

COMBINATION CHEMOTHERAPY IN SOLID TUMORS

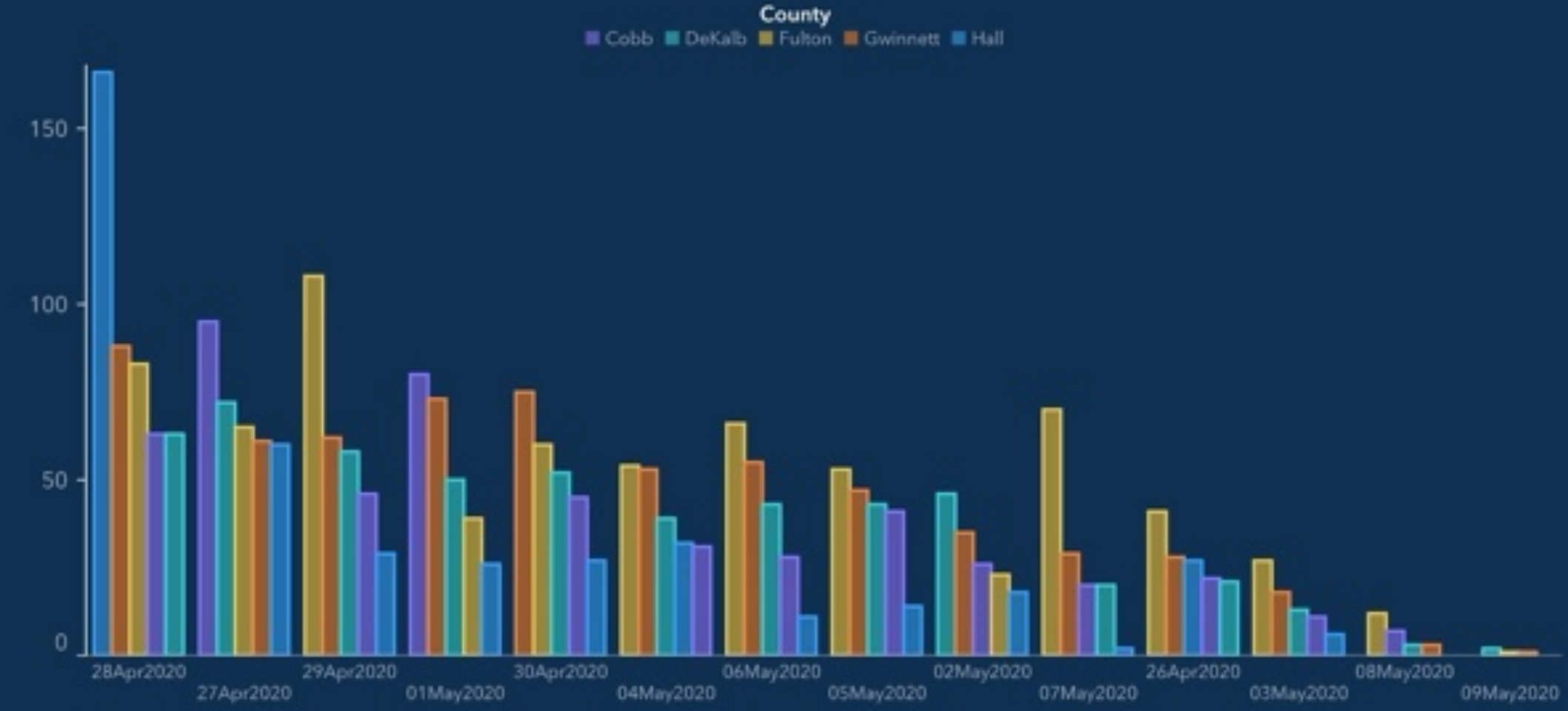


<https://www.tandfonline.com/doi/full/10.1080/17498430.2018.1450055#d1e341>

**BAD CHARTS ARE HARD TO READ,
MUDDLE OR HARM OUR MESSAGE
AND EVEN OUR CREDIBILITY**

Top 5 Counties with the Greatest Number of Confirmed COVID-19 Cases

The chart below represents the most impacted counties over the past 15 days and the number of cases over time. The table below also represents the number of deaths and hospitalizations in each of those impacted counties.



[nature](#) > [technology features](#) > article

TECHNOLOGY FEATURE | 09 December 2024

Bad bar charts distort data – and pervade biology

Scientific ‘shorthand’ could be misrepresenting findings in published papers.

By [Amanda Heidt](#)



Bar charts are ubiquitous in the life-science literature, yet a study suggests that they're often used in ways that can misrepresent research findings. The preprint¹, which was first posted on bioRxiv in September and has yet to be peer reviewed, found that in a collection of nearly 3,400 papers from 2023 that included at least one bar chart, almost one-third distorted the data in some way, highlighting a need for increased data literacy among scientists and for a

Beyond Bar and Line Graphs: Time for a New Data Presentation Paradigm

[Tracey L Weissgerber](#)^{1,*}, [Natasa M Milic](#)^{1,2}, [Stacey J Winham](#)³, [Vesna D Garovic](#)¹

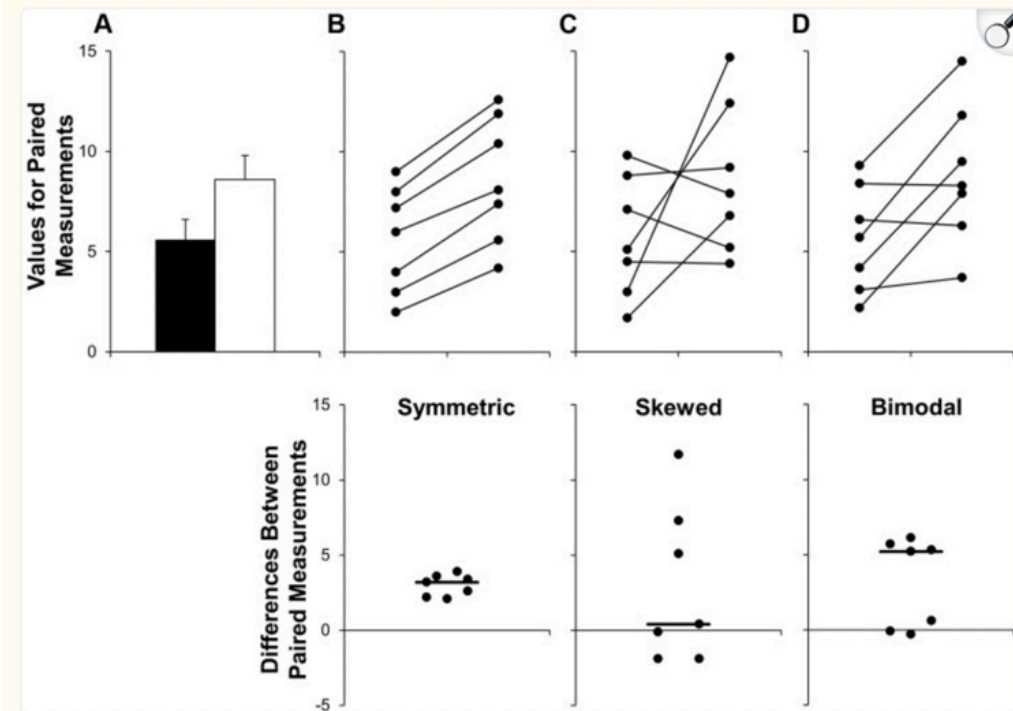
► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#)

PMCID: PMC4406565 PMID: [25901488](#)

Abstract

Figures in scientific publications are critically important because they often show the data supporting key findings. Our systematic review of research articles published in top physiology journals ($n = 703$) suggests that, as scientists, we urgently need to change our practices for presenting continuous data in small sample size studies. Papers rarely included scatterplots, box plots, and histograms that allow readers to critically evaluate continuous data. Most papers presented continuous data in bar and line graphs. This is problematic, as many different data distributions can lead to the same bar or line graph. The full data may suggest different conclusions from the summary statistics. We recommend training investigators in data presentation, encouraging a more complete presentation of data, and changing journal editorial policies. Investigators can quickly make univariate scatterplots for small sample size studies using our Excel templates.

Fig 2. Additional problems with using bar graphs to show paired data.





BASICS OF DATA VISUALIZATION

VISUAL ELEMENTS

position

length

angle

direction

shape

area

volume

saturation/lightness

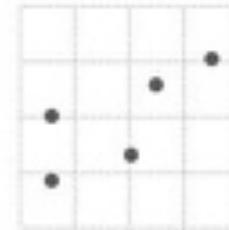
color hue

Visual cues

When you visualize data, you encode values to shapes, sizes, and colors.

Position

Where in space the data is



Length

How long the shapes are



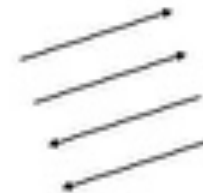
Angle

Rotation between vectors



Direction

Slope of a vector in space



Shapes

Symbols as categories



Area

How much 2-D space



Volume

How much 3-D space



Color saturation

Intensity of a color hue



Color hue

Usually referred to as color



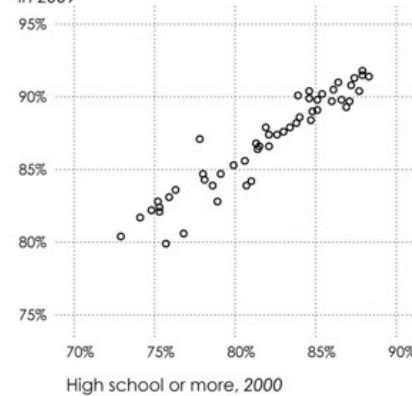
<https://flowingdata.com/data-points/DataPoints-Ch3.pdf>

VISUAL ELEMENTS

combinations

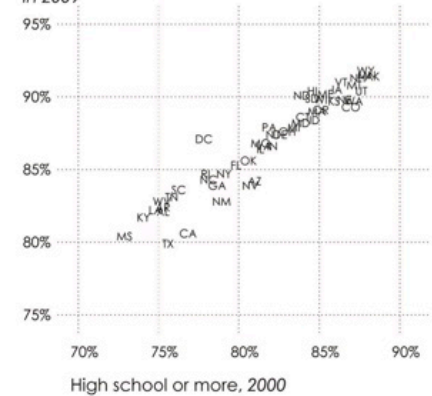
Position

High school
or more,
in 2009



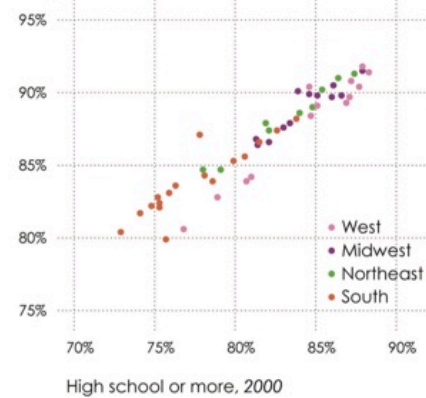
Position + Symbols

High school
or more,
in 2009



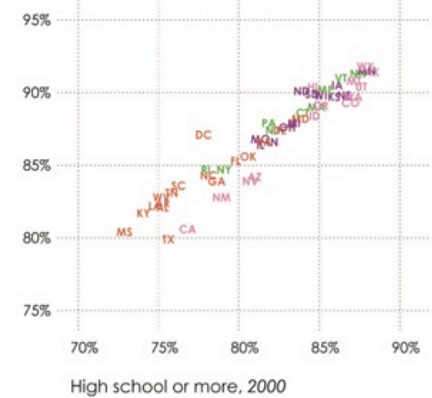
Position + Color

High school
or more,
in 2009



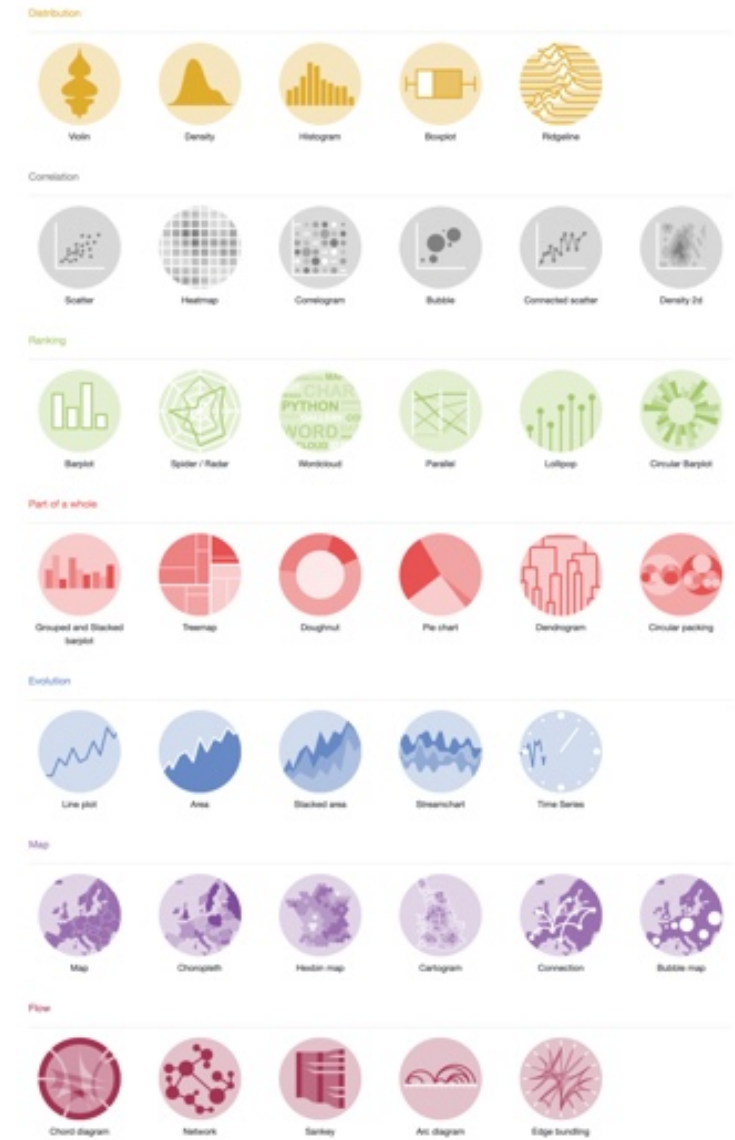
Position + Symbols + Color

High school
or more,
in 2009



<https://flowingdata.com/data-points/DataPoints-Ch3.pdf>

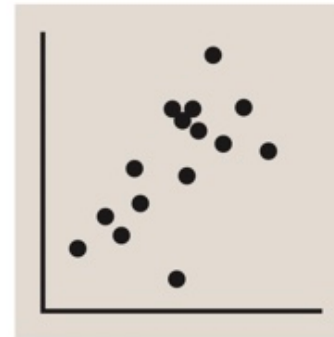
BASIC TYPES OF CHARTS



<https://r-graph-gallery.com>

SCATTERPLOT

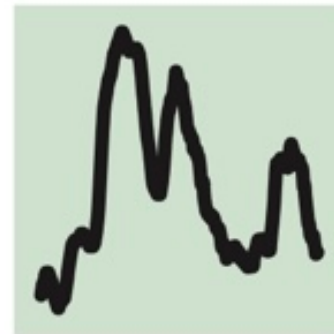
Scatterplot



The standard way to show the relationship between two continuous variables, each of which has its own axis.

LINE CHART

Line



The standard way to show a changing time series. If data are irregular, consider markers to represent data points.

HISTOGRAM

Histogram



The standard way to show a statistical distribution - keep the gaps between columns small to highlight the 'shape' of the data.

BAR CHART

Column



The standard way to compare the size of things. Must always start at 0 on the axis.

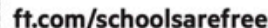
Bar



See above. Good when the data are not time series and labels have long category names.

Read the FT for free

Teachers can register your school and students can create a login by scanning here.

FT FINANCIAL
TIMES

**Royal
Geographical
Society**
with IBG

Advancing geography
and geographical learning

Flow

Show the reader volumes or intensity of movement between two or more states or conditions. These might be logical sequences or geographical locations.

Example 17 uses
 Movement of funds
 journals, information
 graphs.

Sankey

Shows changes in flows from one condition to at least one other, good for tracing the eventual outcome of a complex process.

Waterfall	Designed to show the sequencing of tasks through a flow process, typically budgets, can include +/- components
------------------	--

Chess

 A complex but powerful diagram which can illustrate 2-way flows (and net winner) in a matrix.

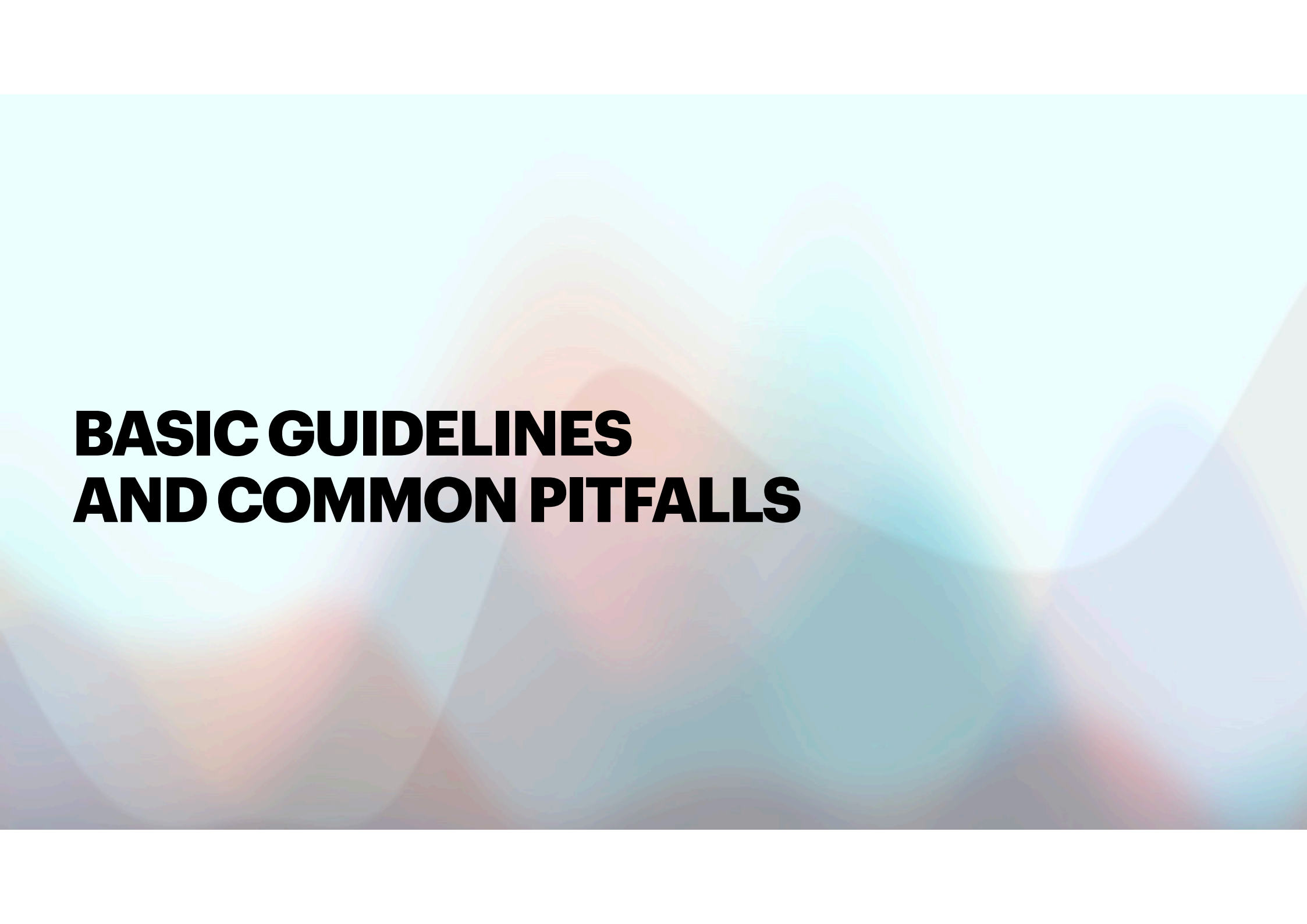
Network



Used for showing the strength and inter-connectedness of relationships of varying types.

**Royal
Geographical
Society**
with IPG

Advancing geography
and geographical learning



BASIC GUIDELINES AND COMMON PITFALLS

Ten guidelines for effective data visualization in scientific publications



Environmental Modelling & Software

Volume 26, Issue 6, June 2011, Pages 822-827



Short communication

Ten guidelines for effective data visualization in scientific publications

Christa Kelleher , Thorsten Wagener

Show more

Share Cite

<https://doi.org/10.1016/j.envsoft.2010.12.006>

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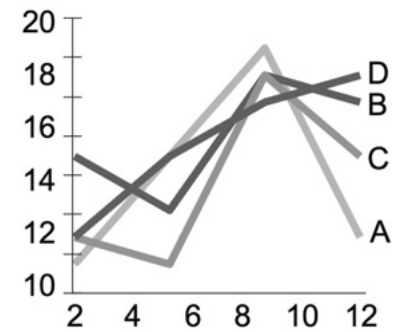
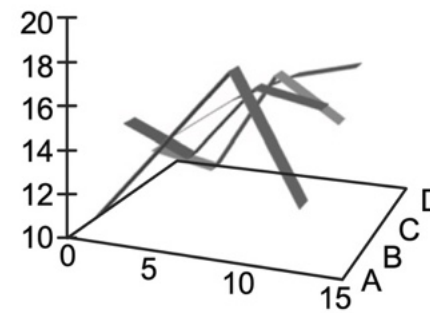
Abstract

Our ability to visualize scientific data has evolved significantly over the last 40 years. However, this advancement does not necessarily alleviate many common pitfalls in visualization for scientific journals, which can inhibit the ability of readers to effectively understand the information presented. To address this issue within the context of visualizing environmental data, we list ten guidelines for effective data visualization in scientific publications. These guidelines support the primary objective of data visualization, i.e. to effectively convey information. We believe that this small set of guidelines based on a review of key visualization literature can help researchers improve the communication of their results using effective visualization. Enhancement of environmental data visualization will further improve research presentation and communication within and across disciplines.

Christa Kelleher, Thorsten Wagener, Ten guidelines for effective data visualization in scientific publications, Environmental Modelling & Software, Volume 26, Issue 6, 2011, Pages 822-827, ISSN 1364-8152, <https://doi.org/10.1016/j.envsoft.2010.12.006>.

GUIDELINE 1

Create the simplest graph that conveys the information you want to convey



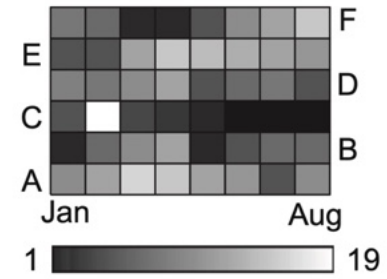
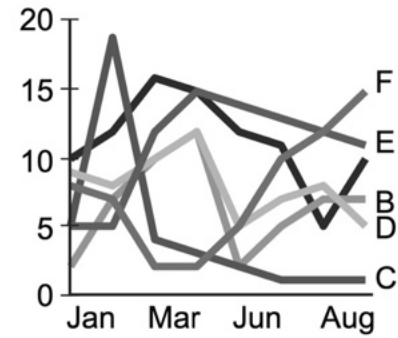
GUIDELINE 2

Consider the type of encoding object and attribute used to create a plot

		Value encoding attribute				
Form	Length		Width		Orientation	
	Size		Shape		Curvature	
	Enclosure		Blur			
Color	Hue		Intensity		Transparency	
Spatial Position	2-D Position		Spatial Grouping		Density	
Motion	Direction		Pathway			

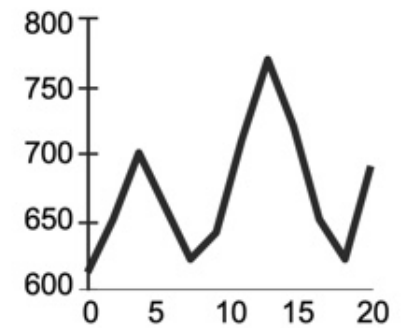
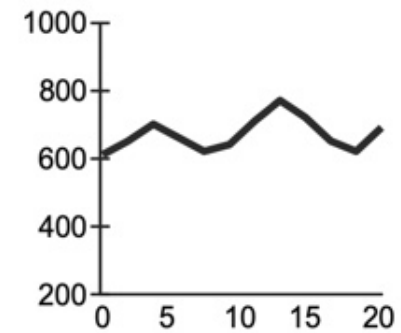
GUIDELINE 3

Focus on visualizing patterns or on visualizing details, depending on the purpose of the plot



GUIDELINE 4

Select meaningful axis ranges





National Review
@NRO



Follow

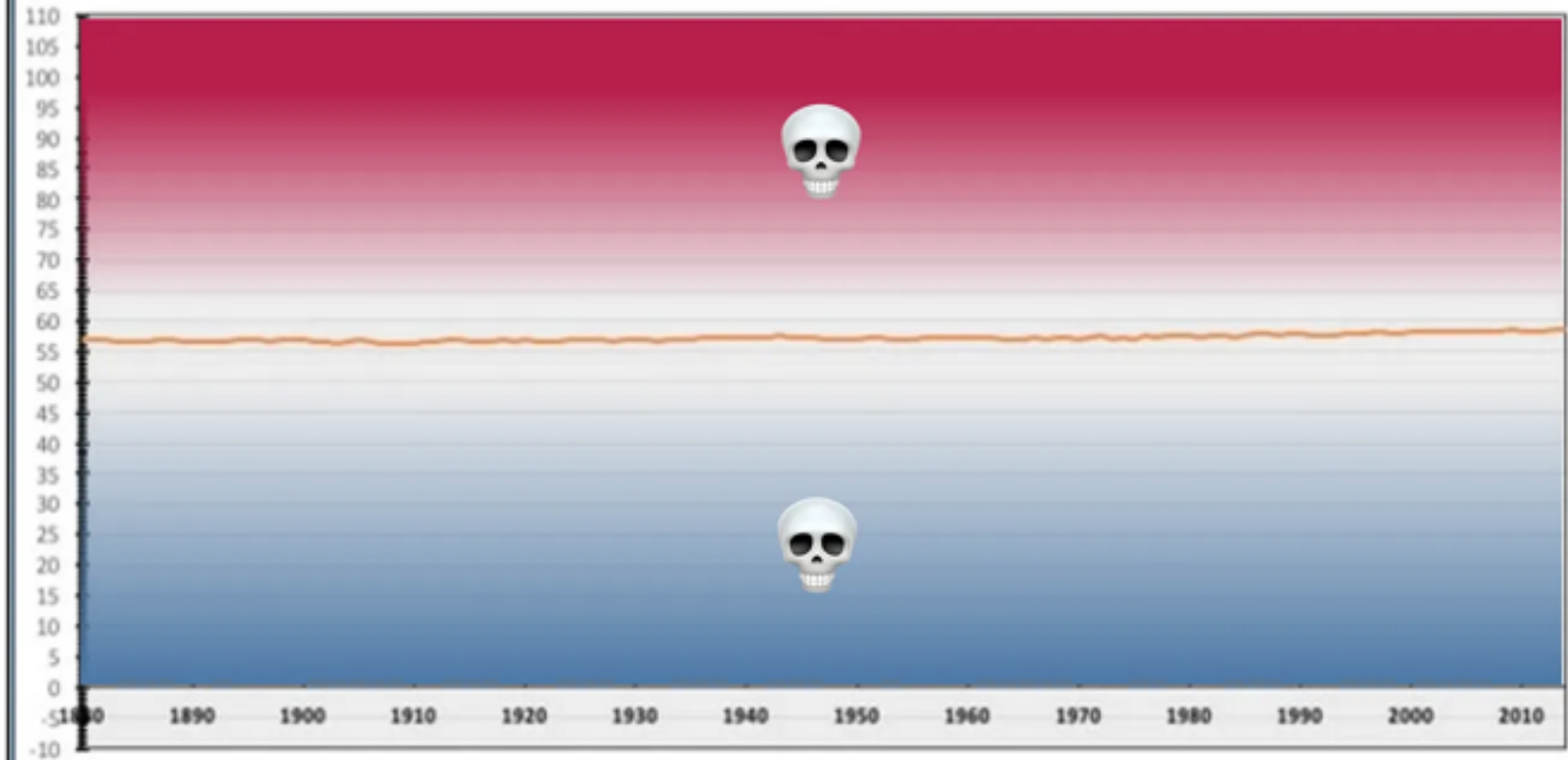
The only [#climatechange](#) chart you need to see. natl.re/wPKpro

(h/t [@powerlineUS](#))



<https://mcorrell.medium.com/an-inconvenient-graph-addc218579e9>

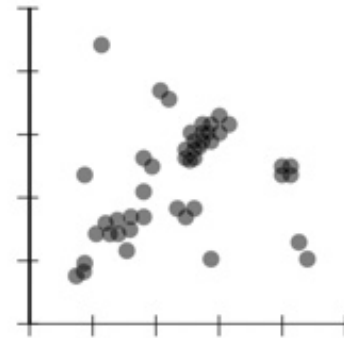
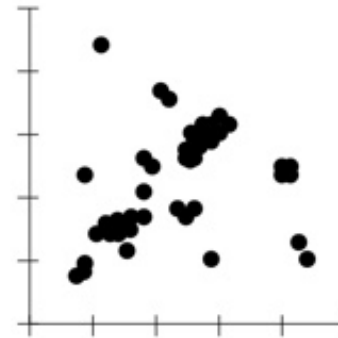
Average Annual Global Temperature in Fahrenheit 1880-2015



<https://mcorrell.medium.com/an-inconvenient-graph-addc218579e9>

GUIDELINE 6

Plot overlapping points in a way that density differences become apparent in scatter plots

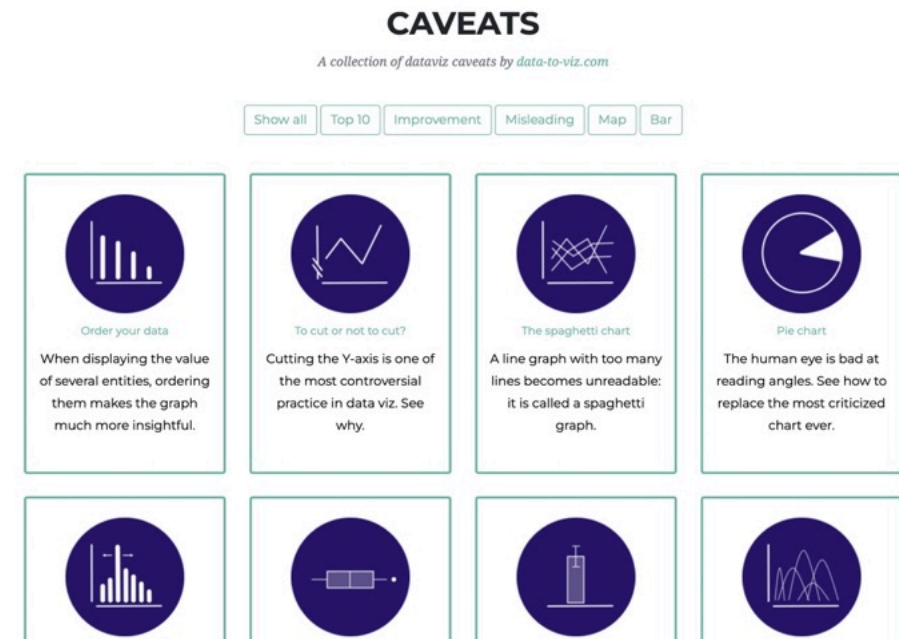


COMMON PITFALLS

and their alternatives

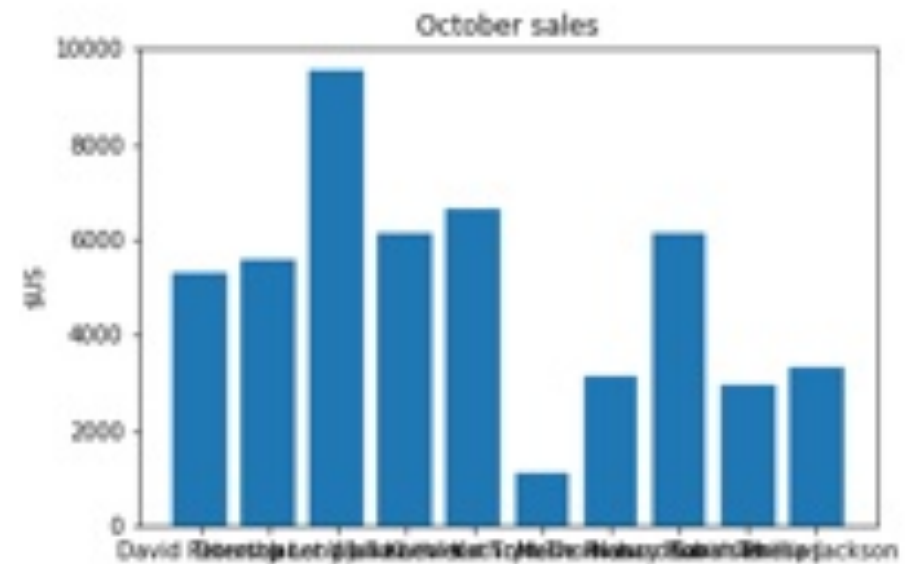
More at

<https://www.data-to-viz.com/caveats.html>



VERTICAL AXIS TEXT

Common problem

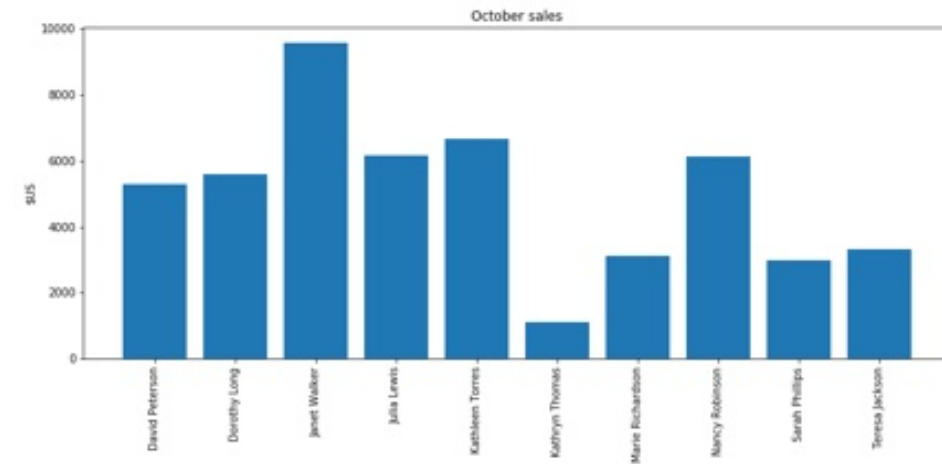


<https://gorelik.net/2017/11/23/how-to-make-a-graph-less-readable-rotate-the-text-labels/>

VERTICAL AXIS TEXT

Solved?

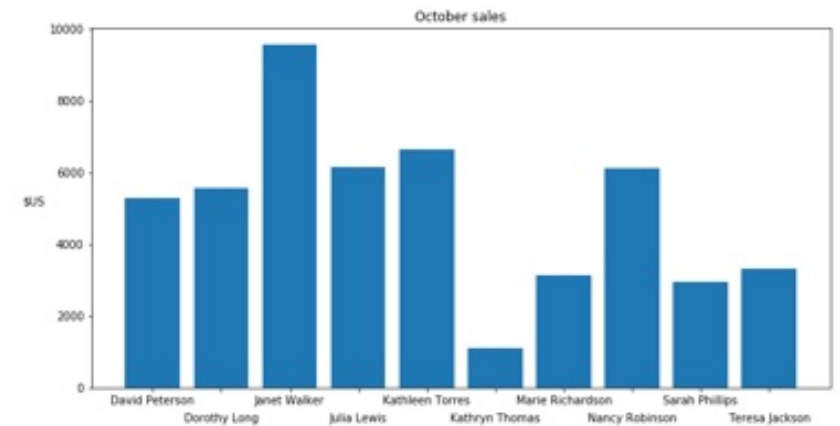
Difficult to read



<https://gorelik.net/2017/11/23/how-to-make-a-graph-less-readable-rotate-the-text-labels/>

VERTICAL AXIS TEXT

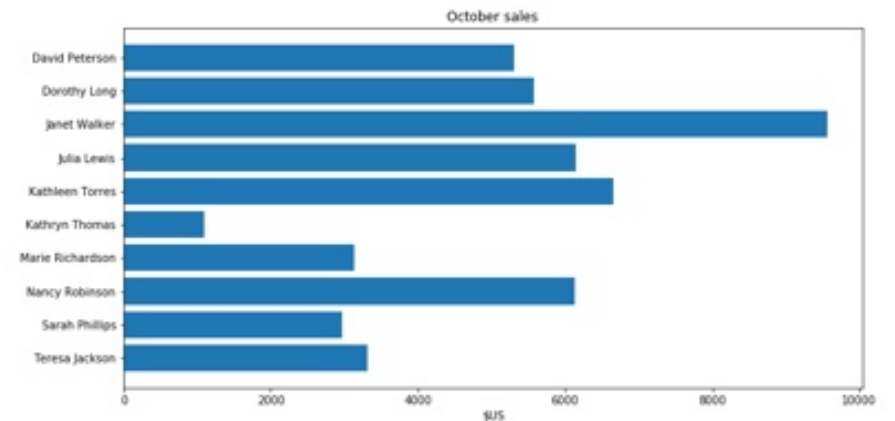
Consider moving the labels



<https://gorelik.net/2017/11/23/how-to-make-a-graph-less-readable-rotate-the-text-labels/>

VERTICAL AXIS TEXT

Rotate the bars

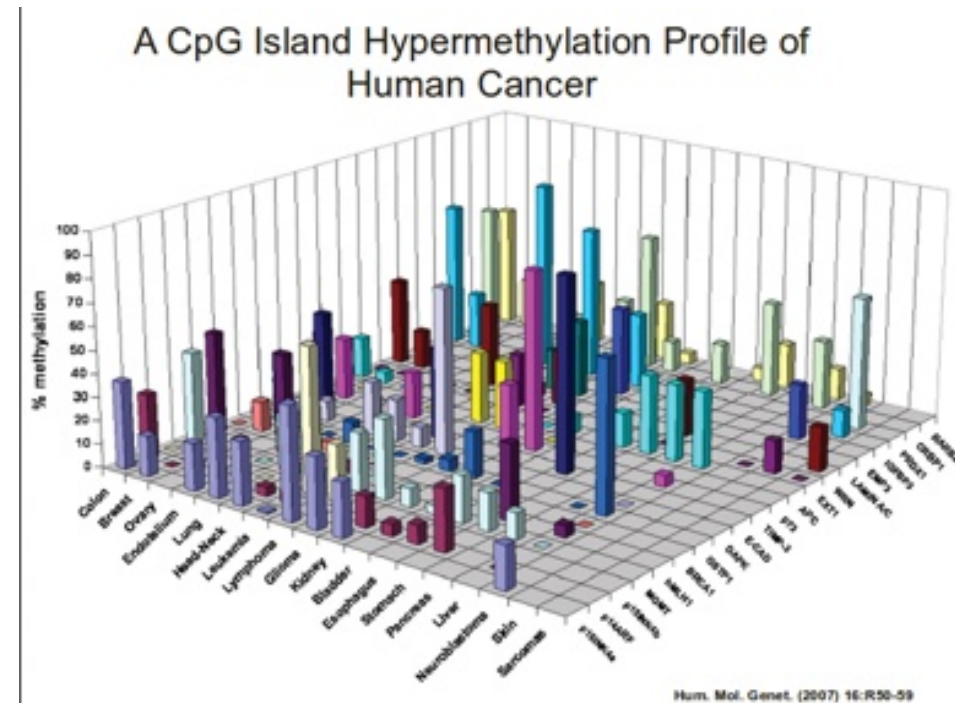


<https://gorelik.net/2017/11/23/how-to-make-a-graph-less-readable-rotate-the-text-labels/>

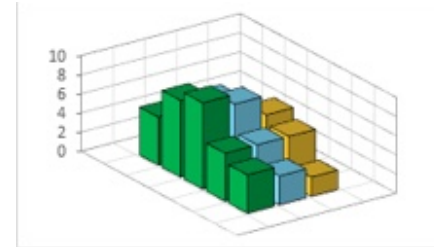
3D

Avoid

Very difficult to read and compare, hidden data

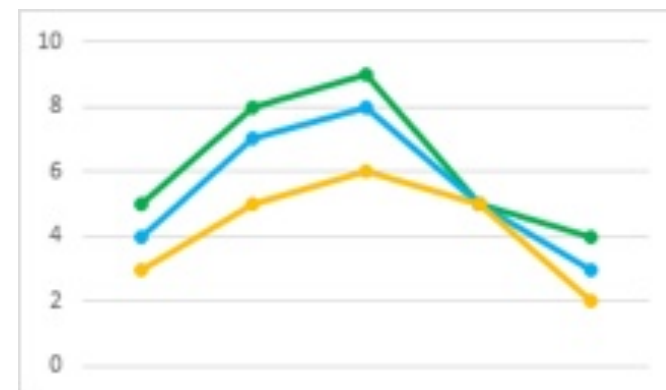
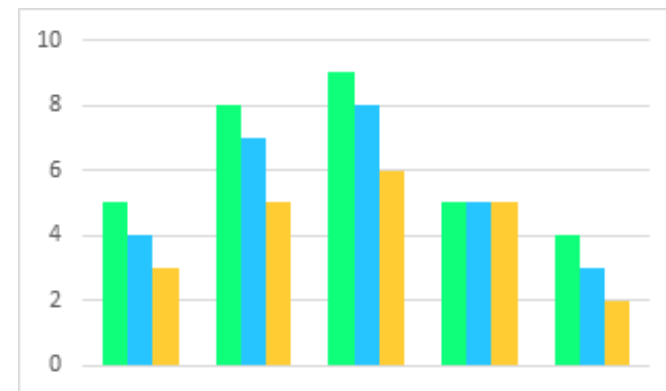


3D



<https://peltiertech.com/3d-bar-chart-alternatives/>

Consider other chart types



PIE CHARTS

Avoid

Perception problems (but not always)

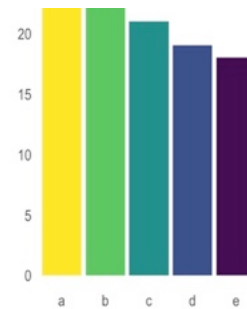
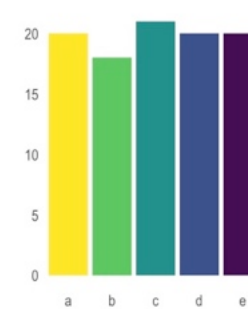
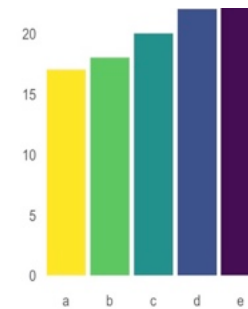
OK for few categories

Can be difficult to read



PIE CHARTS

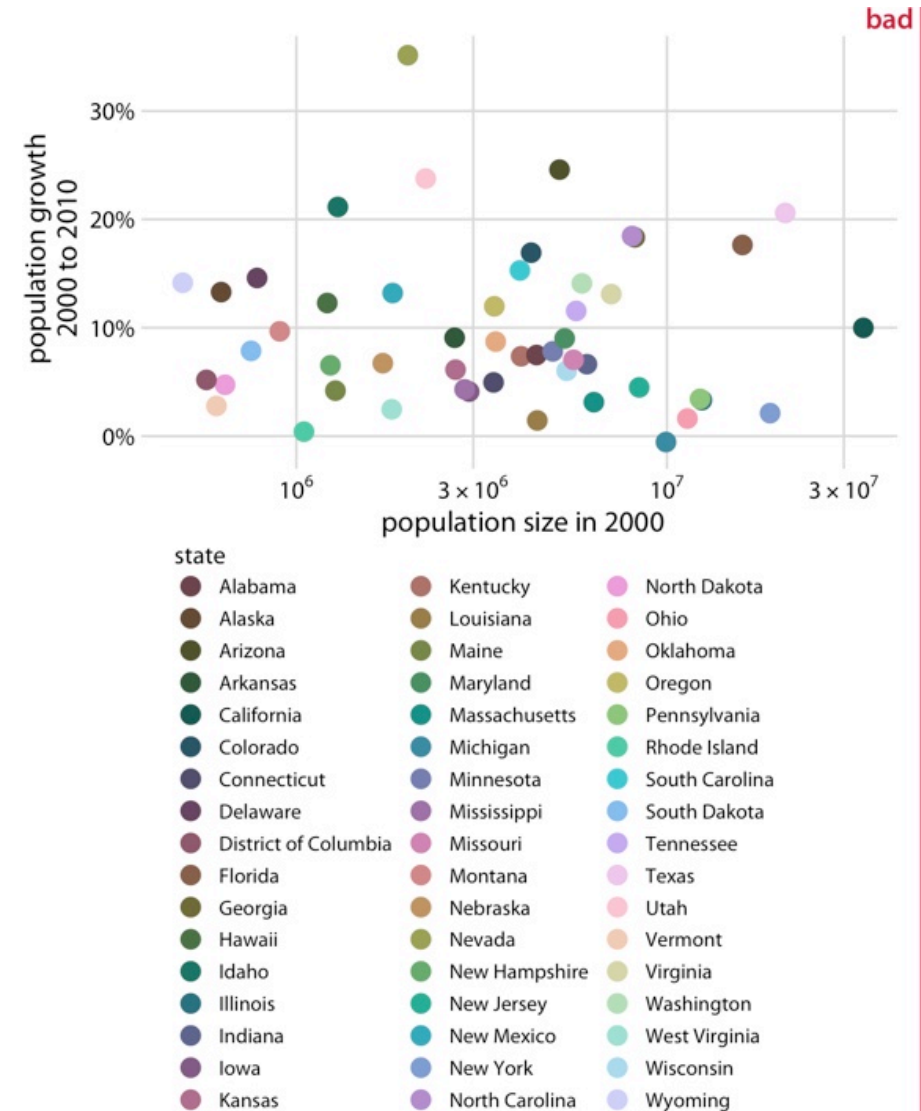
Alternative: bar chart



<https://www.data-to-viz.com/caveat/pie.html>

TOO MANY COLORS

OK to have less than 6-8 categories



<https://clauswilke.com/dataviz/color-pitfalls.html>

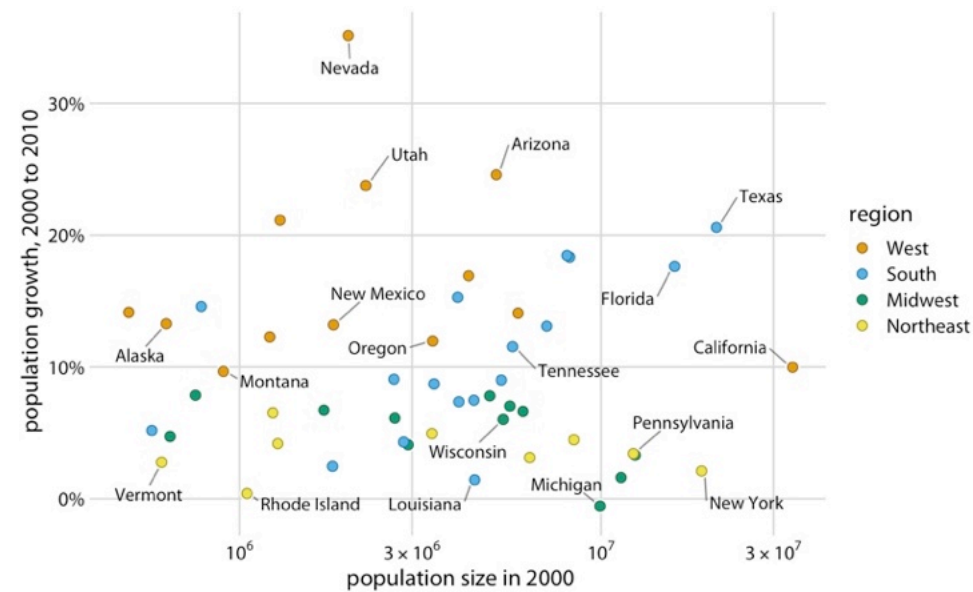
TOO MANY COLORS

Alternatives:

Direct labelling

Bar chart or other chart type

10 ways to use fewer colors in your data visualizations

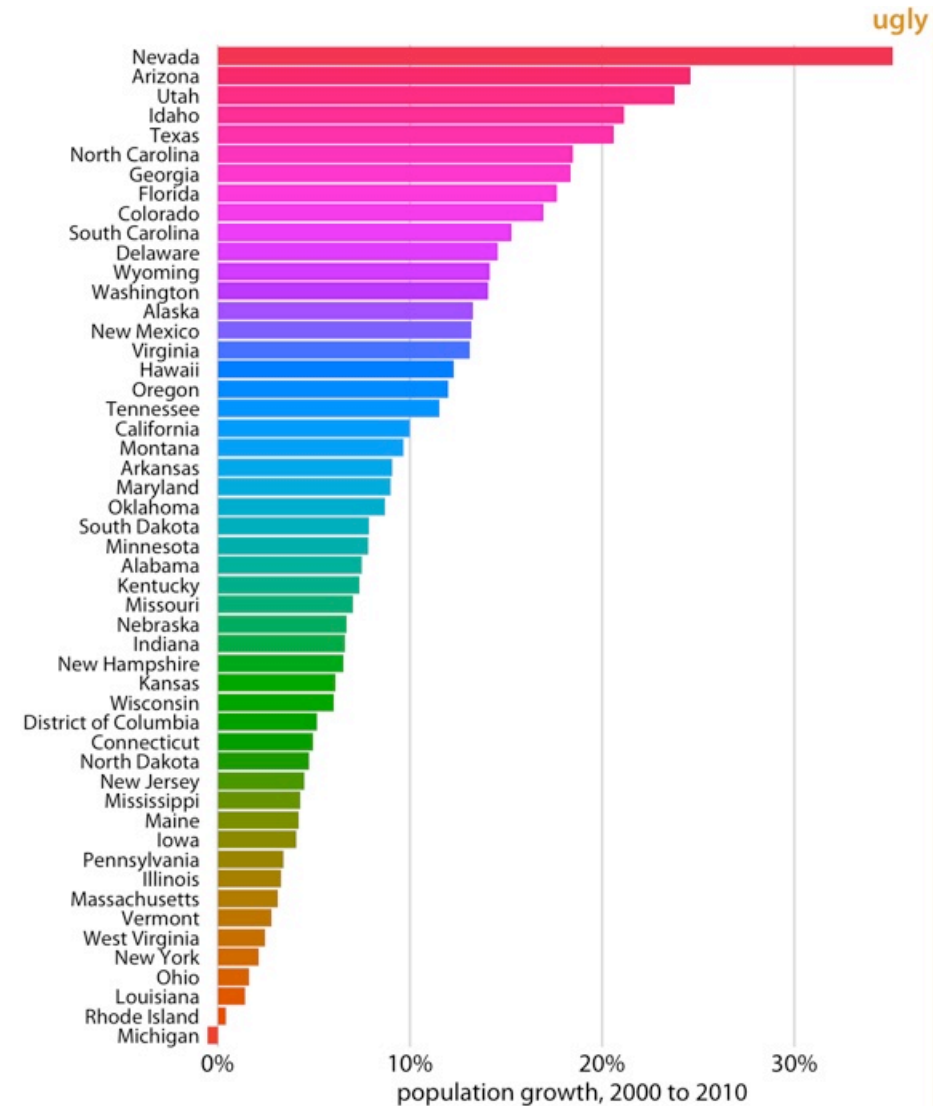


<https://clauswilke.com/dataviz/color-pitfalls.html>

UNNECESSARY COLOR

When color has no information

Use one color instead

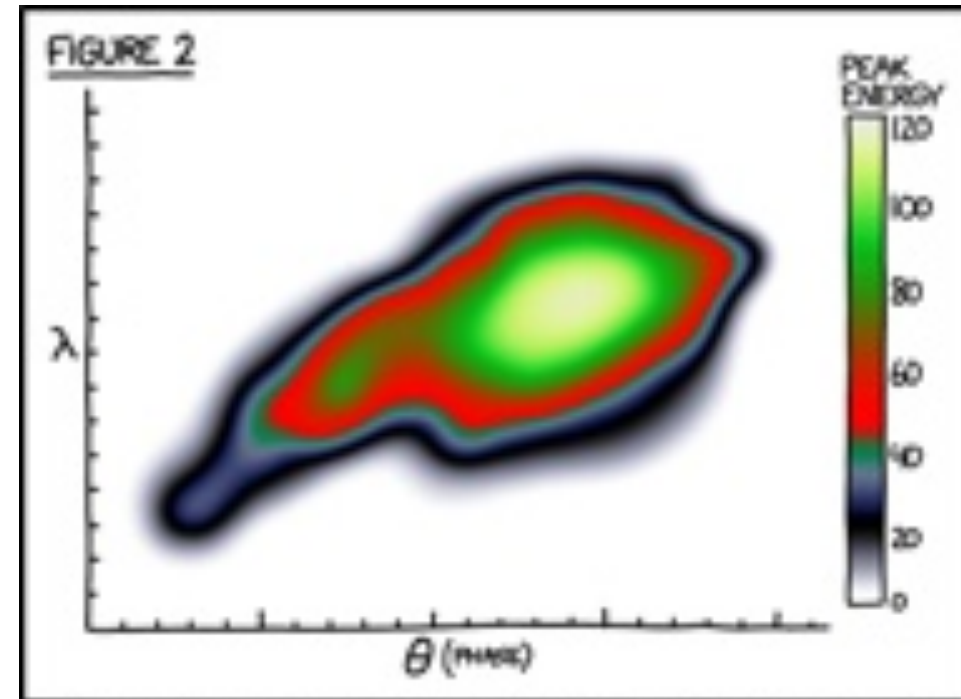


<https://clauswilke.com/dataviz/color-pitfalls.html>

RAINBOW COLOUR SCALES

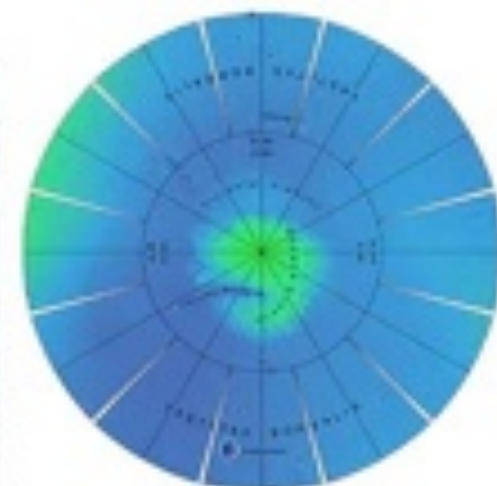
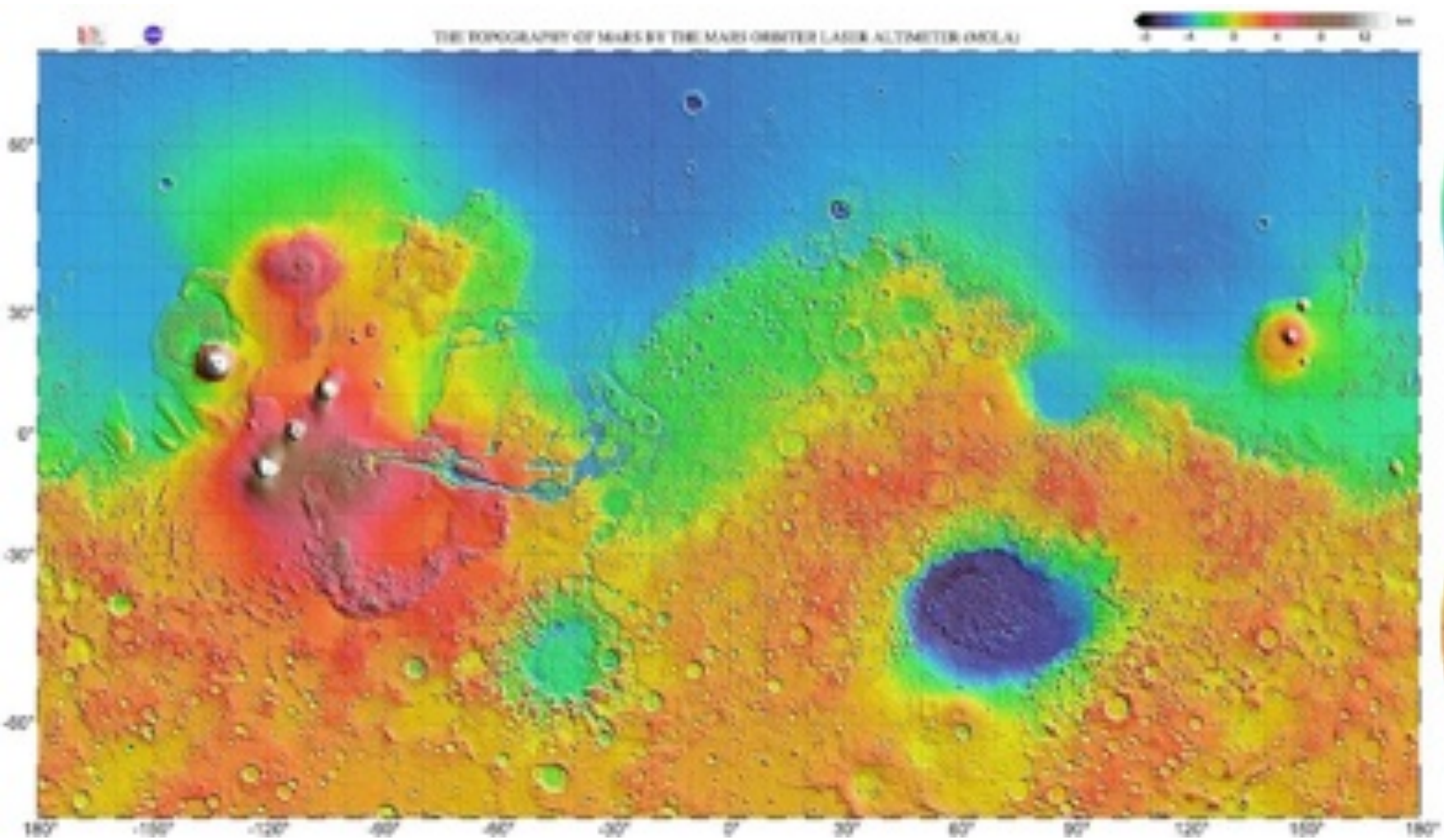
Caution when using the default rainbow colour scales

How rainbow colour maps can distort data and be misleading

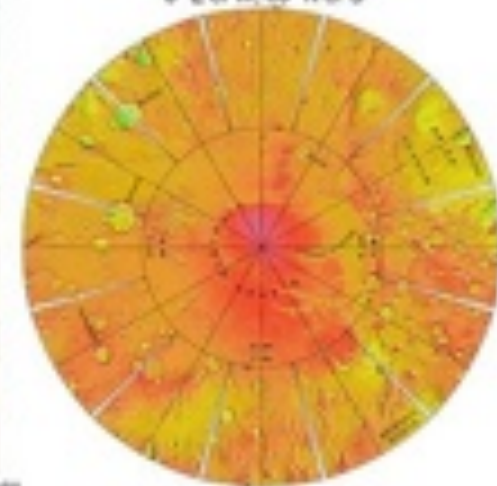


EVERY YEAR, DISGRUNTLED SCIENTISTS COMPETE FOR THE RAINBOW AWARD FOR WORST COLOR SCALE.

<https://xkcd.com/2537/>



0° E or W, 60° N or S

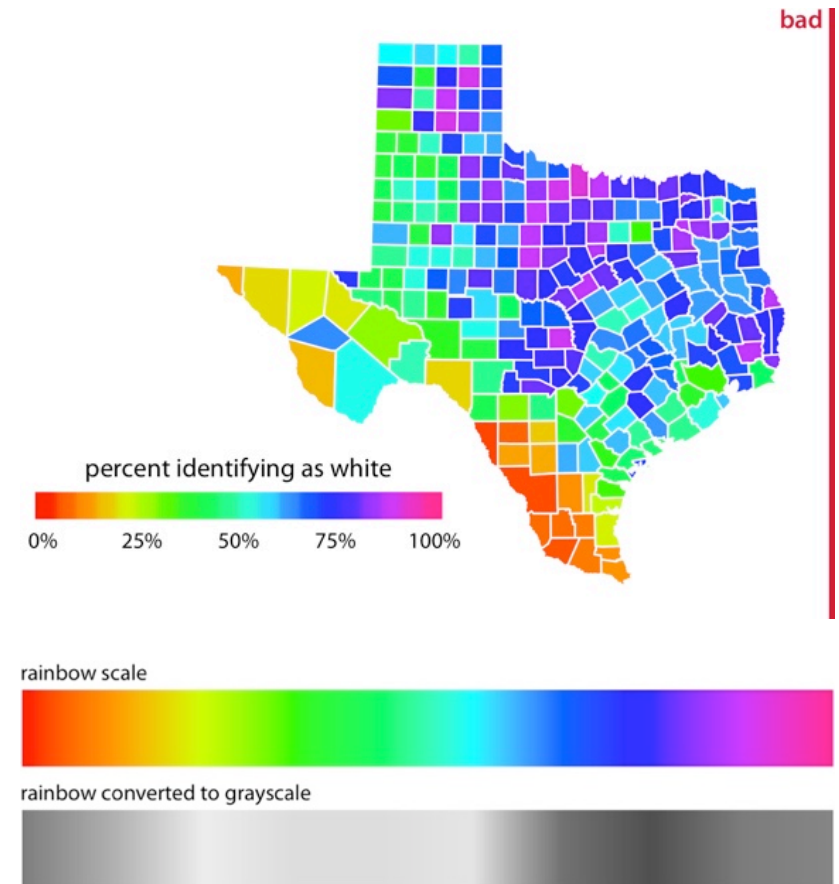


<https://attic.gsfc.nasa.gov/mola/images.html>

RAINBOW COLOUR MAPS

Difficult to compare values

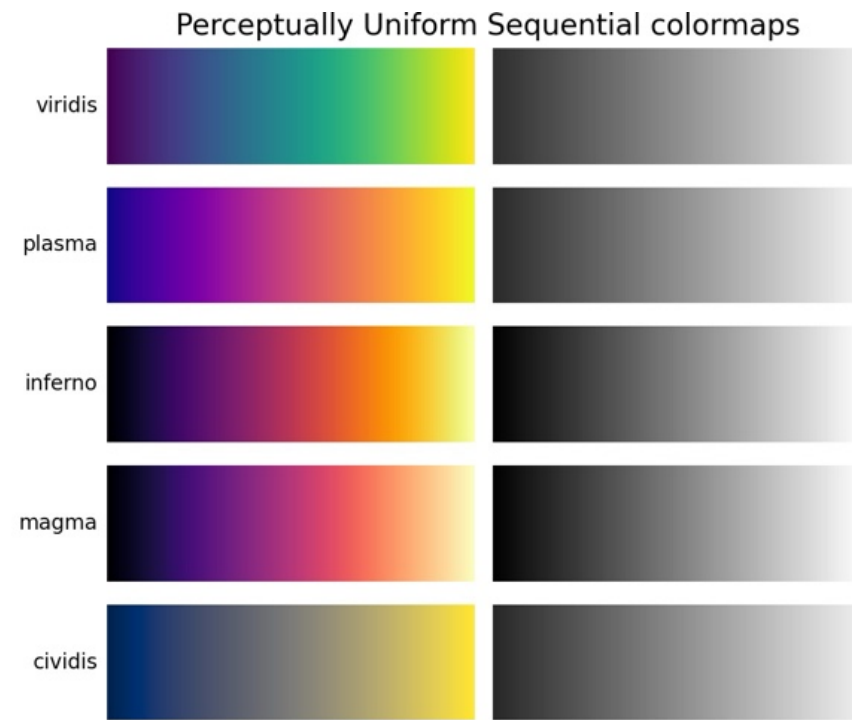
Uneven hues and lightness, misleading



<https://clauswilke.com/dataviz/color-pitfalls.html>

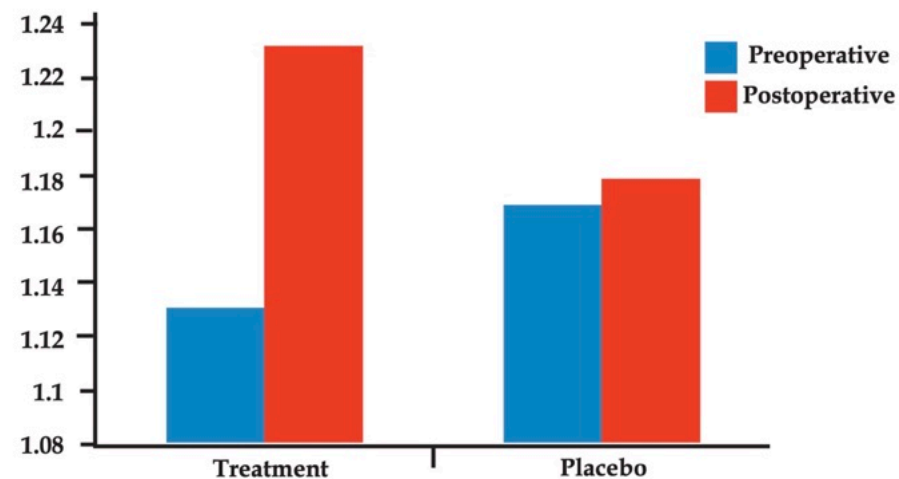
RAINBOW COLOUR SCALES

Use uniform palettes instead



<https://matplotlib.org/stable/tutorials/colors/colormaps.html>

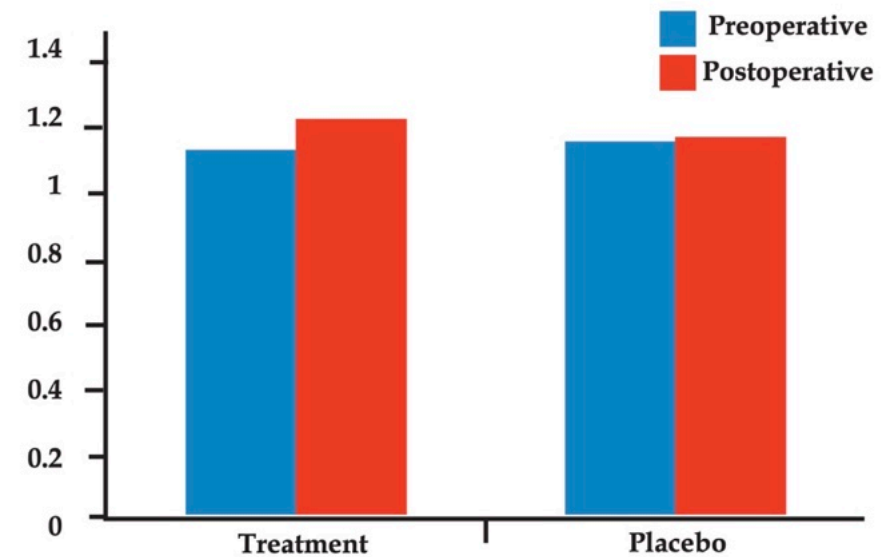
Y-AXIS DOES NOT START AT ZERO



Duquia RP, Bastos JL, Bonamigo RR, González-Chica DA, Martínez-Mesa J. Presenting data in tables and charts. An Bras Dermatol. 2014;89(2):280-5

Y-AXIS DOES NOT START AT ZERO

In bar charts y-axis should start at zero



Duquia RP, Bastos JL, Bonamigo RR, González-Chica DA, Martínez-Mesa J. Presenting data in tables and charts. An Bras Dermatol. 2014;89(2):280-5

Y-AXIS DOES NOT START AT ZERO

No consensus about other charts

Line chart: same trends but small and important changes could be lost

Where to Start and End Your Y-Axis Scale

COMBINED MMR VACCINATION RATE, 1994-5 TO 2014-15, ENGLAND



Source: NHS Immunisation Statistics - England, 2014-15, Table 8 and 9, HSCIC

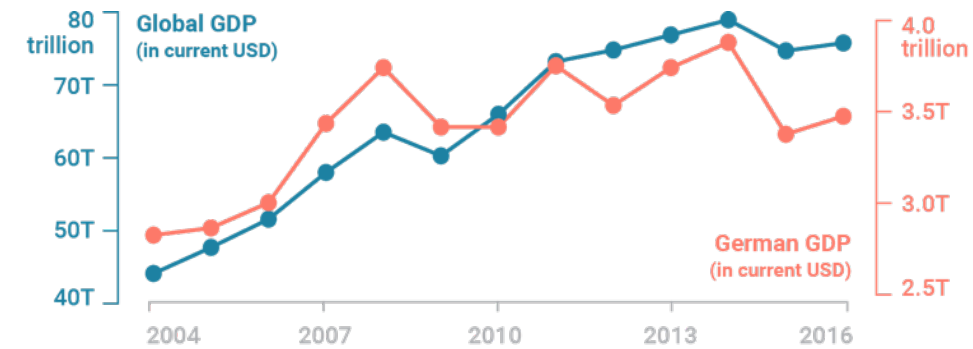


Source: NHS Immunisation Statistics - England, 2014-15, Table 8 and 9, HSCIC

<https://digitalblog.ons.gov.uk/2016/06/27/does-the-axis-have-to-start-at-zero-part-1-line-charts/>

DUAL Y-AXIS

Avoid

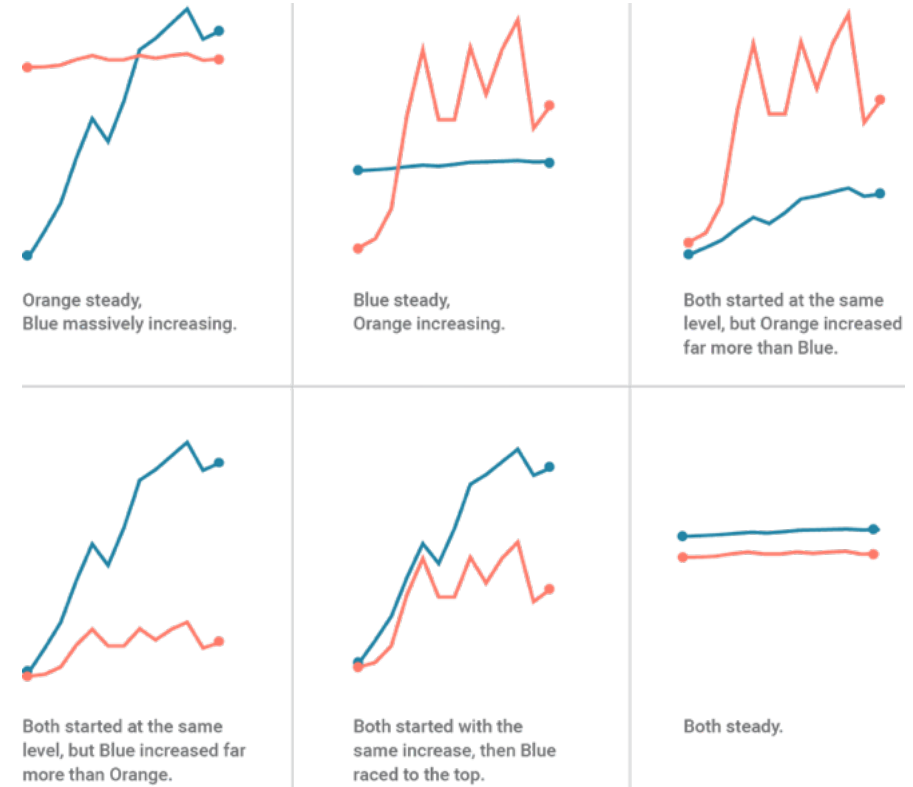


<https://blog.datawrapper.de/dualaxis/>

DUAL Y-AXIS

Difficult to read

Depending on scaling and position, a correlation or other meaning is implied



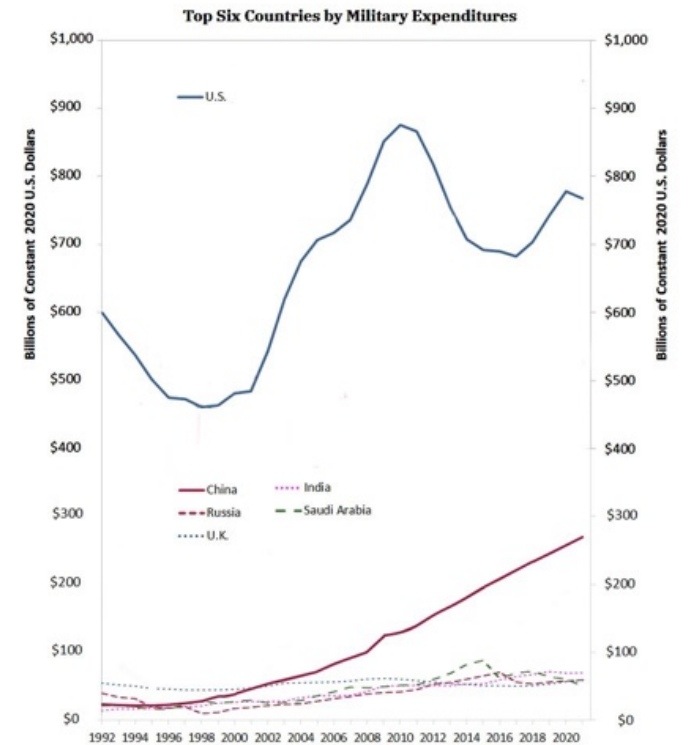
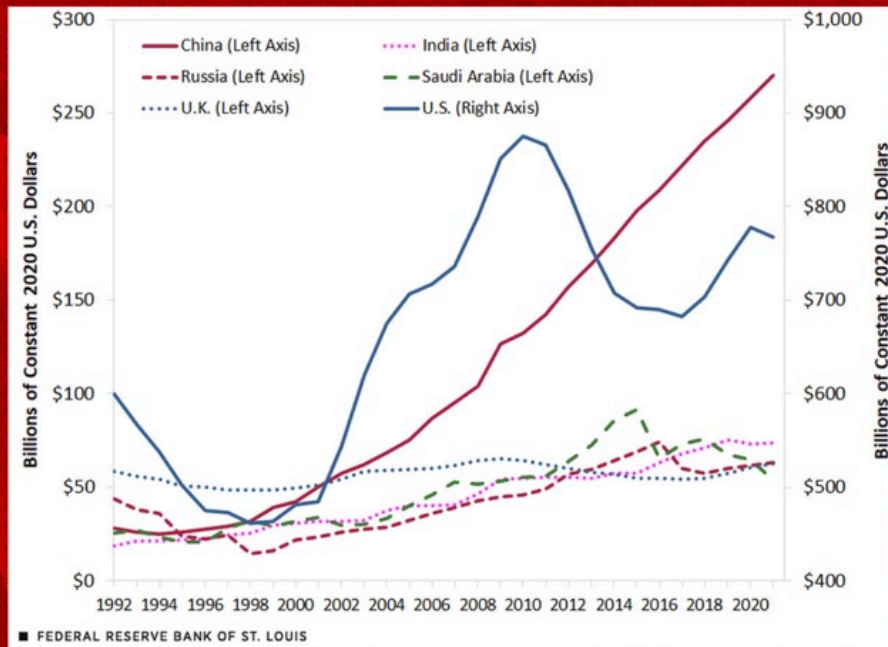
<https://blog.datawrapper.de/dualaxis/>

Exaggerating China's military spending, St. Louis Fed breaks all statistical rules with misleading graph

The Federal Reserve Bank of St. Louis published a jaw-droppingly misleading graph that portrays China as spending more on its military than the US. In reality, the Pentagon's budget is roughly three times larger.



By Ben Norton Published 2023-01-23



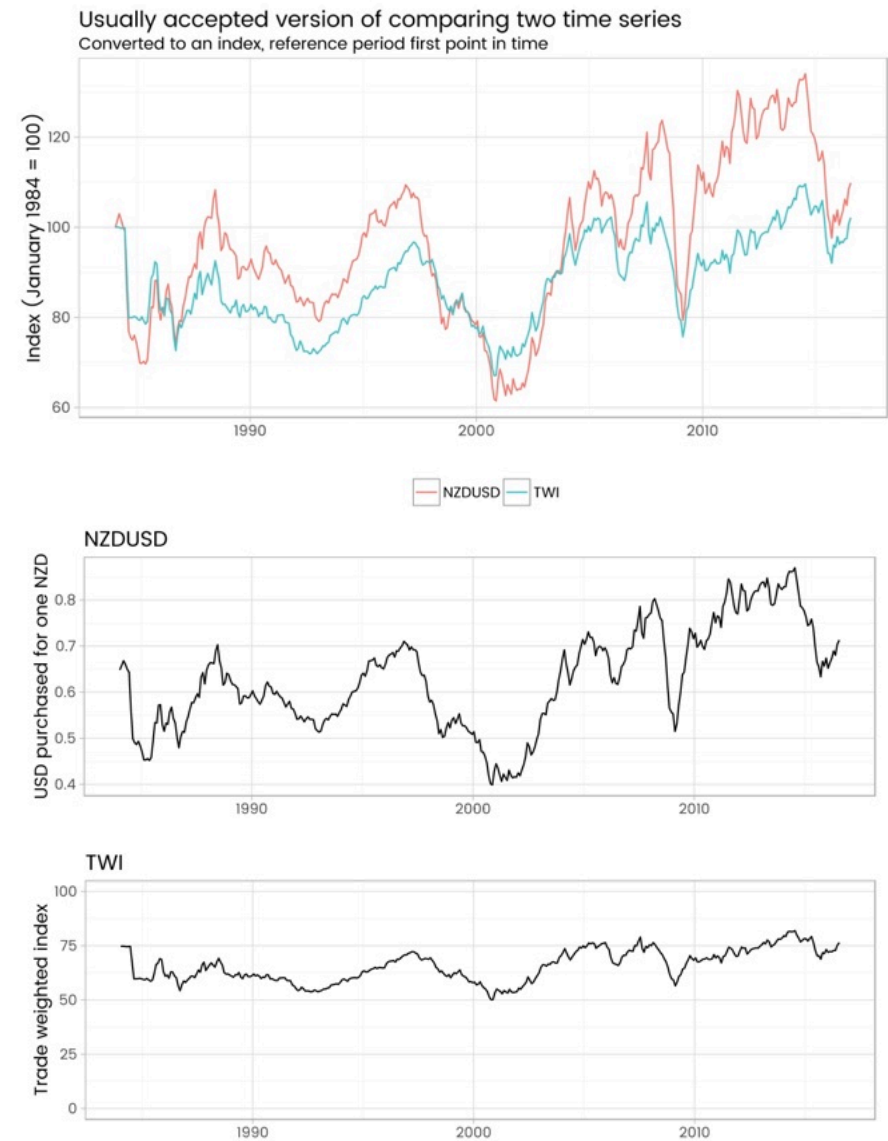
DUAL Y-AXIS

Alternatives:

Indexing

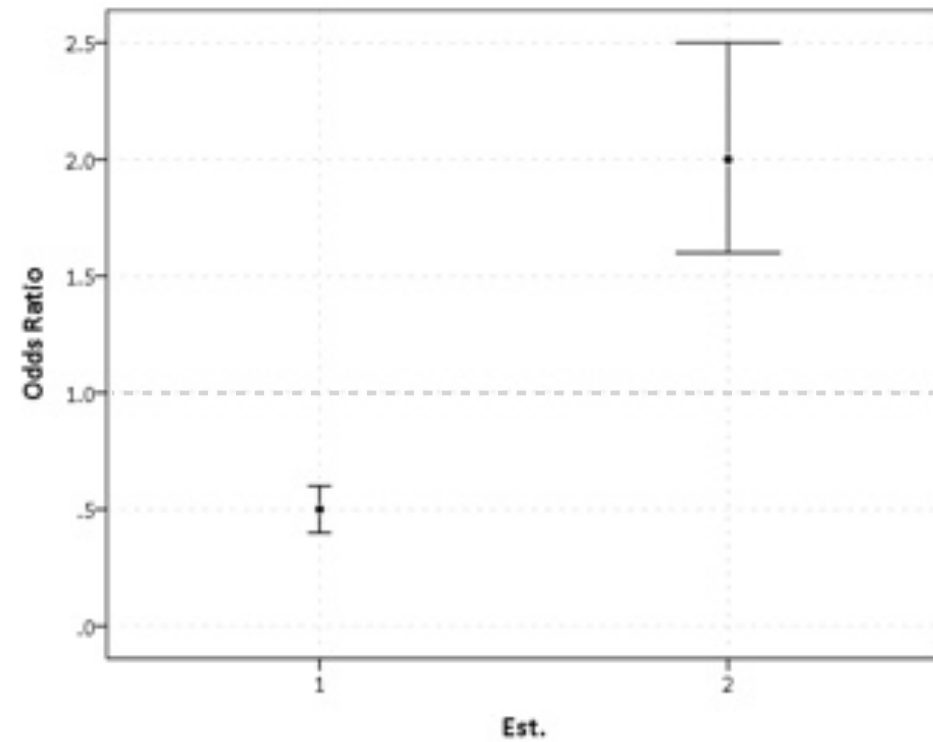
Side by side plots

Why not to use two axes, and what to use instead



<http://freerangestats.info/blog/2016/08/18/dualaxes>

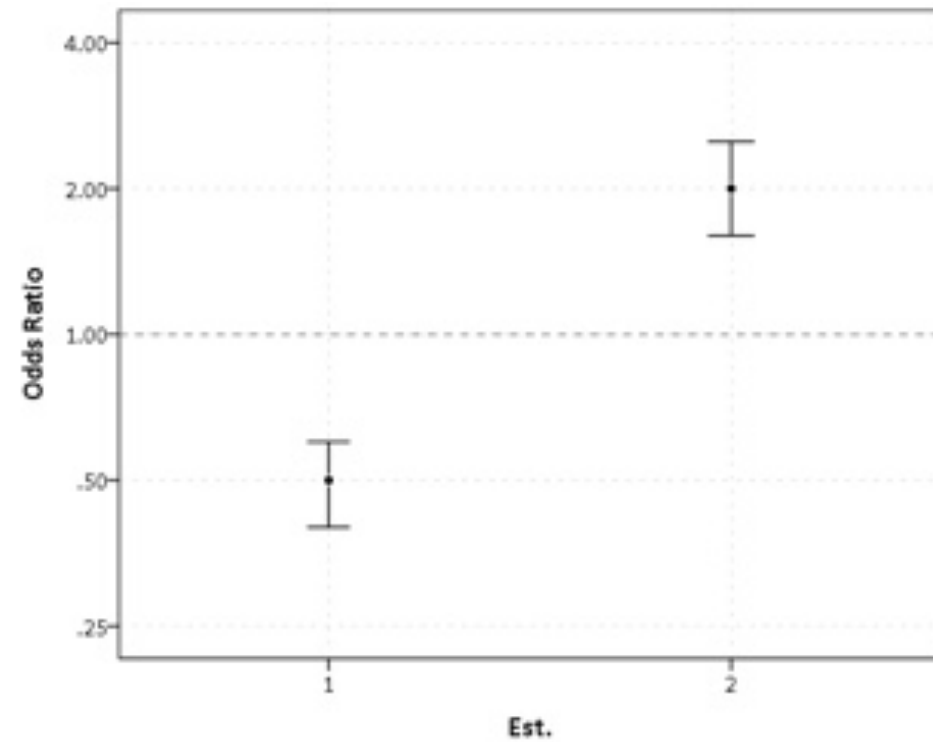
RATIOS THAT DON'T USE LOG AXIS



<https://andrewpwheeler.com/2013/10/26/odds-ratios-need-to-be-graphed-on-log-scales/>

RATIOS THAT DON'T USE LOG AXIS

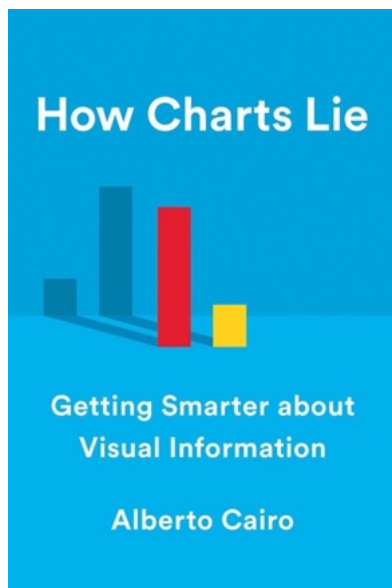
Use logarithmic scale



<https://andrewpwheeler.com/2013/10/26/odds-ratios-need-to-be-graphed-on-log-scales/>

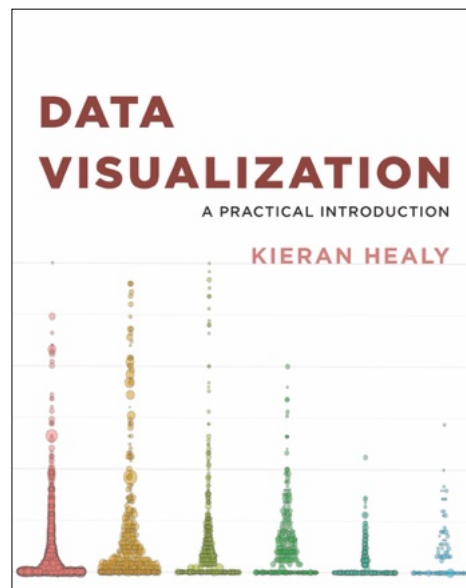


RESOURCES



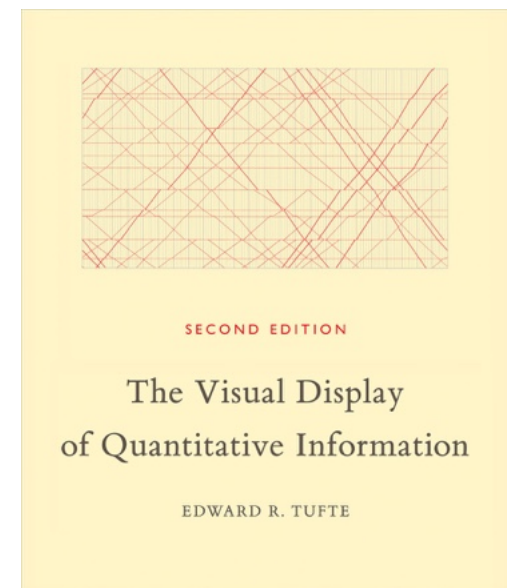
How charts lie: getting
smarter about visual
information

[UU library](#)



Data visualization: a practical
introduction

[UU library](#)
[Free online version](#)



The Visual Display of
Quantitative Information

[UU library](#)

Datawrapper (theory and how-tos, showcase, stories)

Nightingale (tutorials, stories, industry-related news)

FlowingData (showcase)

Defense Against Dishonest Charts
(interactive examples)

The R Graph Gallery (tutorials with code)

A ggplot2 tutorial for beautiful plotting in R (tutorial with code)

Why scientists need to be better at data visualization

Friends Don't Let Friends Make Bad Graphs



FIRST, LET'S REVIEW COMMON MYTHS in DATA VIZ

MYTH:
LINE GRAPHS are
for CONTINUOUS
DATA ONLY

The LINES that CONNECT the POINTS HAVE to MAKE SENSE

Example:
SURVEY DATA
SCOPEGRAPH
(a line graph w/
only 100 points)

EMPLOYEE SATISFACTION



MYTH:
BARS are
ALWAYS
BETTER

BARS are a GOOD PLACE to START... but NOT ALWAYS the BEST
ASK "what do I want my audience to see?"



Try other types of charts and decide what meets your needs

MYTH:
GRAPHS *
MUST have
a ZERO
BASELINE

* THIS IS TRUE for BAR CHARTS



MYTH:
PIE CHARTS
are EVIL

When USING a PIE, ASK yourself WHY?



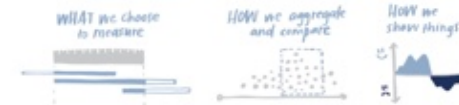
Studies prove that people read pies & charts by comparing AREA, not ANGLE

If you think pies make sense for your data & audience, test them out to see!

MYTH:
UNBIASED
DATA EXISTS

We are BIASING our DATA at EVERY STEP of the PROCESS

RULE! DON'T LIE with DATA



MYTH:
MORE DATA is
ALWAYS BETTER

BEFORE CHASING after MORE DATA, ask "WHAT will it HELP us DO or DECIDE?"

Audience & context are important when it comes to the right amount of data

MYTH:
AVERAGES
ALWAYS WORK to
SUMMARIZE DATA

YOU NEED to UNDERSTAND the DISTRIBUTION, SPREAD, and VARIABILITY



Averages can mislead by hiding a spread in a single number

MYTH:
THERE is a SINGLE
RIGHT ANSWER
when VISUALIZING
DATA

YOU SHOULD ALWAYS CONSIDER when SHOWING DATA: WHAT is your GOAL?



**DO NOT
DECEIVE!**



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 <https://karaman.is>

 <https://github.com/gkaramanis>

 <https://bsky.app/profile/karaman.is>



PAUS

GGPLOT2 WORKSHOP

bit.ly/isr-ggplot2

<https://gkaramanis.github.io/ISR-Workshop>

ISR ggplot2 Workshop

Georgios Karamanis